When Is Label-Free Adaptation Knowable? Helpful, Harmful, and Unknowable Regimes Under Distribution Shift

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Abstract—**Test-time adaptation (TTA) improves models on unlabeled target data, yet the very unlabeled objective that recovers accuracy under one distribution shift can silently degrade it under another—and, without labels, a deployed system cannot tell which is happening. Existing methods propose increasingly robust *adaptation mechanisms*; we instead study a prior question: *can a system decide, without labels, whether adaptation should happen at all?* We formalize adaptation benefit as the excess risk between an adapted candidate and a frozen baseline conditioned on label-free observable evidence, and separate test-time regimes into *knowably helpful*, *knowably harmful*, and *unknowable*. We prove (Theorem 1, with an explicit witness) that when two target worlds induce the same observable evidence but opposite benefit, no label-free rule can certify the correct decision in both, so the worst-case-optimal action on such evidence is to abstain. Under an observable risk-alignment assumption we give a finite-sample adapt/freeze/abstain certificate (Theorem 2) that controls the false-adapt and false-freeze rates at a chosen level, and we identify a provable positive regime (Corollary 1, covariate shift). Our central theoretical result is an exact benefit-sign *frontier*: $\operatorname{sign}\Delta$ is identifiable from label-free observables if and only if an observable margin $|M|$ exceeds the calibration-drift budget β on the region where the two models disagree—the drift is the sole irreducible obstruction, and prior label-free heuristics (ATC, agreement-on-the-line, AETTA) are recovered as its degenerate $\beta{=}0$ face. We then close the question the frontier leaves open with a *testability dichotomy*: no label-free-checkable assumption class can identify the benefit sign—an unobservable swap flips every sign while moving no observable of any order—so the minimal untestable supplement is exactly *one bit*, which provably suffices under a falsifiable structural hypothesis whose budgets become auditable (falsifiable at level α, never verifiable) and whose label-free rate matches the labeled one at $m^{-1/2}$. The theory is exactly five theorems. We instantiate the framework (KGA) on a 123-task anomaly-routing benchmark and a controlled mixed-regime study: KGA abstains where the true benefit is near zero, refuses a fusion path that hurts on 80% of tasks (cutting regret-to-oracle $\approx 11\times$ versus always-adapt), and in the mixed regime beats an always-freeze policy by ≈ 0.10 AUROC. In *online non-stationary* CIFAR-10 test-time adaptation, across a helpful-dominated and a harm-dominated stream, no fixed policy is robust—always-adapt collapses in the harm regime (0.339) and always-freeze forgoes gains in the helpful regime—while KGA is near-best in both, giving the highest mean accuracy across regimes (0.490 vs $0.456/0.444$). On a *pre-registered* CIFAR-10-C stress grid (severity$\times$batch$\times$composition$\times$aggressiveness; harmful base rate 0.16–0.34), KGA *beats both* trivial policies for Tent and EATA—regret-to-oracle 0.0016 vs always-adapt 0.009 and always-freeze 0.12, at a 0% false-adapt rate—and ties the collapse-resistant SAR. At ImageNet scale the harmful-candidate win survives a *mechanism-faithful* SAR re-implementation—KGA *beats both* trivial policies on the SAR stream (regret-to-oracle 0.023 vs. $0.112/0.027$), confirming it is not an implementation artifact—while it ties near-oracle always-adapt for the benign Tent/EATA candidates. On two *natural* shifts (WILDS Camelyon17, ImageNet-R) the harmful regime is real and brackets the detectability frontier (harmful base rate $0.29/0.14$, label-free harm-AUC $0.78/0.66$), but at our current (frozen, source-calibrated) operating point the certificate *abstains*—correctly where harm is undetectable, conservatively where it is detectable—so we report these as regime-existence evidence, not a natural-shift policy win. Under that frozen, Camelyon-free $\tau^\star$ KGA does *not* beat both on Camelyon17, and recalibrating $\tau^\star$ on the external panel itself would forfeit its validation role—so the honest route is a scale-invariant τ, not per-dataset retuning. We are explicit about scope: where harmful adaptation is mild, always-adapt is already strong, so the certificate's distinctive value is bounded by how often *catastrophic, detectable* harm occurs—itself the quantity the theory characterizes.**

Index Terms—**test-time adaptation, distribution shift, selective prediction, identifiability, label-free risk estimation**

Scope. We report both positive and negative results and do not claim universal improvement; all numbers come from code in the public repository (§XI). A detailed proof/experiment status note appears at the end of §I.

I. Introduction

Deployed models meet distribution shift, and TTA promises to use unlabeled target data to recover lost accuracy. But adaptation is double-edged: entropy minimization that sharpens a good boundary under mild covariate shift can collapse to a degenerate solution under label shift; a reliability gate that rescues a failed sensor can dilute a strong detector on clean data. The field has answered with sturdier mechanisms (sample selection, sharpness-aware updates, resets). We ask the prior question:

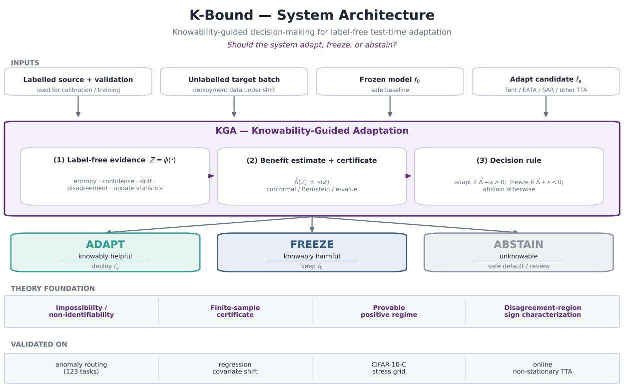

Fig. 1. K-Bound system architecture. Inputs (labelled source/validation, an unlabelled shifted target, a frozen f_0, a candidate adaptation f_a) feed the KGA pipeline—label-free evidence $Z = \phi(\cdot)$, a finite-sample certificate $\widehat{\Delta}(Z) \pm \varepsilon(Z)$, and a decision rule—which routes to one of three regimes: ADAPT (knowably helpful), FREEZE (knowably harmful), or ABSTAIN (unknowable). The theory layer (Thms 1–5) justifies the trichotomy; RGF is the worked reliability-gated-fusion instantiation that K-Bound generalises.

> *Can observable test-time evidence separate helpful, harmful, and unknowable regimes of label-free adaptation, yielding a computable rule for when a system should adapt, freeze, or abstain?*

We treat adaptation as a decision constrained by *identifiability*, not only as an optimization problem.

a) Contributions.: (1) **The exact benefit-sign frontier** (§V, our main result): sign Δ is identifiable from label-free observables *iff* an observable margin $|M|$ exceeds the calibration-drift budget β on the disagreement region (Theorem 3); the drift γ is the *sole, irreducible* obstruction (Cor. 1), and prior label-free heuristics (ATC, agreement-on-the-line, AETTA) are its degenerate β=0 face (Thm. 3(iv)); (2) a knowability formulation of label-free adaptation (§III–IV); (3) an impossibility theorem with an explicit witness establishing the unknowable regime (Thm. 1); (4) a finite-sample adapt/freeze/abstain certificate with false-adapt control (Thm. 2); (5) a provable positive regime and an open conjecture for the general case (Cor. 1; resolved by Thm. 4); (6) KGA and real experiments on anomaly routing, deep TTA, ImageNet-scale corruptions, and two natural shifts (Camelyon17, ImageNet-R) (§VII); and (7) an explicit, honest scope (§VIII).

A per-result proof-and-validation status summary is in Appendix A; the honest empirical scope (where the certificate's value is large versus where always-adapt already suffices) is stated in §VIII.

II. Related work and positioning

We position K-Bound as a precise delta over five adjacent lines of work. In each case the distinction is the one the theory formalizes: between *improving* an unlabeled objective and *certifying the sign* of its benefit before acting.

A. Test-time adaptation methods

Entropy-based TTA updates a frozen model on unlabeled target data. **Tent** [1] minimizes prediction entropy over the batch-norm affine parameters; **SHOT** [2] combines information maximization with pseudo-labeling for source-free adaptation; **TTT** [3] trains a self-supervised auxiliary task and updates it at test time; **EATA** [4] adds entropy filtering of unreliable samples plus a Fisher penalty against forgetting; and **SAR** [5] adds sharpness-aware updates with a reset that aborts incipient collapse. These methods make adaptation *more robust*, and several detect or suppress instability, but all commit to adapting: none emits a *pre-adaptation* certificate separating helpful, harmful, and unknowable evidence, and none can *abstain* on the adapt/freeze decision itself. K-Bound is mechanism-agnostic—it *wraps* any such candidate (we evaluate Tent/EATA/SAR).

B. Protected and guarded adaptation

A second line guards adaptation *during* the update. **POEM** [33] protects TTA by online entropy matching—a betting-style monitor that halts or dampens a single failure mode—and **Schirmer et al.** [6] run anytime-valid risk monitors over the TTA stream, raising a sequential alarm when adaptation degrades a proxy risk; both descend from sequential harmful-shift tracking for deployed models (**Podkopaev & Ramdas** [7]). These are the nearest neighbours of our certificate, and the delta is precise: they act *reactively, during or after* the update and control a false-*alarm* rate in one failure direction, whereas K-Bound decides *before* committing, poses the *three-way* decision with an explicit *unknowable* region (where abstention is mandatory, Cor. 1), controls the false-*adapt* rate on the benefit sign itself, and proves the matching impossibility floor (Prop. 1). The two are complementary: a pre-adaptation certificate can gate what a runtime monitor then tracks.

C. Label-free accuracy and error estimation

A third line predicts target *performance* without labels: **ATC** [8] thresholds softmax confidence calibrated on source; **Agreement-on-the-Line** [10], [9] exploits the linear correlation between in- and out-of-distribution model *agreement* to extrapolate accuracy; **AETTA** [12] estimates TTA accuracy from dropout-prediction disagreement. Closest in spirit, **Kim et al.** [11] use agreement-on-the-line to decide *when* TTA helps and to select TTA hyperparameters without labels—a genuine decision use of an accuracy estimate, but a plug-in one: it is exactly the β=0 face of our frontier (Thm. 3(iv)), sound only when calibration transfers, with no finite-sample false-adapt control and no abstain region. Estimating $R_T(f)$—or detecting that TTA has failed *post hoc*—is strictly different from certifying sign $\big(R_T(f_0) - R_T(f_a)\big)$ *before* adapting with a controlled false-adapt rate, the object K-Bound bounds.

D. Unsupervised risk estimation and identifiability

The hardness of our problem is inherited from unsupervised risk estimation. **Steinhardt & Liang** [13] estimate risk without labels only under a conditional-independence structure; **Ben-David et al.** [14] bound target risk by source risk plus an $\mathcal{H}$-divergence; **Lipton et al.** [15] (BBSE) correct label shift under an invertible confusion matrix; **Rosenfeld & Garg** [16] bound target error under an explicit, unverifiable assumption (their Remark 3.9 asserts informally the impossibility our Theorem 4 proves); the rank-one agreement algebra underlying our Theorem 4 is classical [17], [18], [20], [21], its dependent-classifier diagnostics appear in [19], and label-free algebraic evaluation with consistency alarms is developed in the NTQR program [22]. Each shows risk is identifiable only under structural assumptions and impossible in general—precisely the boundary our Theorem 1(ii) makes quantitative. We weaken the target from a *risk* to the *sign of a difference* of risks, which is identifiable on a strictly larger set (the observable disagreement region, Lemma 1).

E. Selective prediction, conformal, and anytime-valid inference

Selective prediction [24] abstains on individual *predictions*; we instead abstain on the *adaptation decision*. Our finite-sample certificate reuses split-conformal prediction (**Vovk et al.** [25]; **Angelopoulos & Bates** [26]) to build the radius ε, and its streaming form (Appendix A) is a testing-by-betting e-process in the safe-anytime-valid-inference tradition (**Shafer** [27]; **Ramdas et al.** [28]); Vovk's conformal test martingales [29] embody, for exchangeability, the falsify-not-verify stance our budget audit (Thm. 4(iv)) takes for the drift budget.

The surviving gap. No prior method provides a single adapt/freeze/abstain certificate with an *explicit, proven* unknowable region and a false-adapt guarantee. Table I makes the gap precise.

F. Benchmarks, code, and reproducibility

TTA progress is benchmarked on standardized corruption suites—CIFAR-10/100-C and ImageNet-C (Hendrycks & Dietterich) and the RobustBench common-corruptions track—and the field increasingly emphasizes standardized, executable evaluation (RobustBench; DomainBed). Most TTA methods ship official code, but evaluations vary in batch regime, stream composition, and update budget—*precisely* the axis along which adaptation flips between helpful and harmful. We therefore release K-Bound as a *pre-registered, executable* artifact: a public repository with pinned environments, a one-command reproduction harness, auto-downloaded standard corruption data, fixed seeds, machine-checked theorem validators, and the full stratified result manifests behind every table and figure (§XI). Our certificate builds on split-conformal prediction (Vovk et al.; Angelopoulos & Bates) and, for the streaming variant, testing-by-betting e-processes (Shafer; Ramdas et al.).

III. Problem setup

Source $P_S(X, Y)$ has labels at train/validation; target $P_T(X, Y)$ exposes only $P_T(X)$ at test time. Fix a loss ℓ, a frozen model f_0, and an adapted/gated candidate f_a. Target risk $R_T(f) = \mathbb{E}_{P_T}[\ell(f(X), Y)]$. The *adaptation benefit* is

$$\Delta = R_T(f_0) - R_T(f_a), \qquad \Delta > 0 \iff \text{adapting helps.} \tag{1}$$

Observable evidence $Z = \phi(X_{1:m}, f_0, f_a)$ is any statistic computable *without* target labels (entropy, confidence, score-distribution drift, detector disagreement, predicted-class balance, update norm, density-ratio diagnostics). The conditional benefit is $\Delta(z) = \mathbb{E}[\ell(f_0(X), Y) - \ell(f_a(X), Y) \mid Z = z]$. The central question is when sign $\Delta(z)$ is recoverable from Z alone.

a) Notation.: f_0 is the frozen model and f_a the adapted/gated candidate; $\Delta = R_T(f_0) - R_T(f_a)$ is the population adaptation benefit and $\Delta(z)$ its value conditioned on evidence $Z{=}z$; $\widehat{\Delta}$ is a label-free estimator of Δ with confidence radius ε at miscoverage level α; $g(Z) \in \{\text{ADAPT}, \text{FREEZE}, \text{ABSTAIN}\}$ is the decision rule; and $D = \{x : f_0(x) \neq f_a(x)\}$ is the observable disagreement region. *Regret-to-oracle* denotes $R(g) - R(g^\star)$ against the per-condition oracle $g^\star = \mathbf{1}[\Delta > 0]$.

IV. Definitions

Definition 1 (Observable risk alignment)**.** Z is *risk-aligned* if it identifies or bounds sign $\Delta(z)$ (it need not reveal the label, only whether adaptation helps).

Definition 2 (Regimes)**.** At evidence z: *knowably helpful* if Z certifies $\Delta(z) > 0$ (**adapt**); *knowably harmful* if Z certifies $\Delta(z) < 0$ (**freeze**); *unknowable* if the same Z is compatible with both signs (**abstain**).

What "label-free" means here. *Target-label-free* (our setting): no labels from the target/test stream are ever used. *Validation labels allowed*: labeled source and a held-out labeled validation/probe set may calibrate the benefit estimator $\widehat{\Delta}$ and the radius ε. *Target-unsupervised evidence*: the deployment batch contributes only X, predictions, entropy, confidence, drift and other label-free statistics Z. The online-TTA results that use a clean labelled validation probe (§VII) are target-label-free in this sense—they use no *target* labels—not unsupervised of all labels.

V. Theory: five theorems

The theory is exactly five theorems: the impossibility floor (Thm 1), the operational certificate (Thm 2), the exact knowability frontier (Thm 3), the one-bit dichotomy that closes the paper's open conjecture (Thm 4), and the label-free rate (Thm 5). Supporting variants (multi-class/regression reductions, label-shift and drift boundaries, the ambiguity-reach unification, router capacity, anytime certificates, the γ-meter, and the agreement–accuracy

Work	label-free?	adapts?	predicts benefit *before* adapting?	abstains on the decision?	false-adapt guarantee?
Tent	yes	yes	no	no	no
SHOT	yes	yes	no	no	no
EATA	yes	yes	partial (filtering)	no	no
SAR	yes	yes	stability only	no	no
AETTA	yes	monitors	post-hoc estimate	—	no
KGA (ours)	yes	wrapper	**yes**	**yes**	**yes**

TABLE I
POSITIONING. PRIOR TTA ADAPTS AND (SOME) DETECT FAILURE; NONE GIVES A *PRE-ADAPTATION* ADAPT/FREEZE/ABSTAIN CERTIFICATE WITH A FALSE-ADAPT BOUND.

law) are corollaries of these five and are developed, with full proofs and their own validators, in the companion manuscript; this section gives statements, proof sketches, and the validation hooks. Every numeric claim traces to a script in the public repository.

Lemma 1 (Reduction to the disagreement region)**.** *Let $D = \{x : f_0(x) \neq f_a(x)\}$ (observable). For binary Y and 0/1 loss, with $\bar{a} = \mathbb{P}_T(f_a{=}Y \mid D)$, $\Delta = 2\,\mu(D)(\bar{a} - \frac{1}{2})$, so for $\mu(D) > 0$: $\operatorname{sign}\Delta = \operatorname{sign}(\bar{a} - \frac{1}{2})$. Writing s for any source-calibrated score of f_a, $M := \mathbb{E}_{\mu|D}[s] - \frac{1}{2}$ (observable) and $\gamma := \mathbb{E}_{\mu|D}[\eta_a - s]$ (unobservable), $\boxed{\operatorname{sign}\Delta = \operatorname{sign}(M+\gamma)}$. For K-class 0/1 loss $\Delta = \mu(D)(p_a - p_0)$, and for squared loss the sign reduces to one moment on D; both extend the identity verbatim (manuscript, App. C).*

Proof sketch. Off D the losses coincide; on D exactly one of f_0, f_a is correct (binary), giving $\mathbb{E}[\delta \mid D] = 2\bar{a} - 1$; linearity splits $\bar{a} - \frac{1}{2} = M + \gamma$. Validated: identity exact over 2×10^4 draws (`val_frontier.py`); multiclass to 10^{-16} over 4000 trials (`val_thm5_multiclass.py`). □

Lemma 2 (Plug-in regret identity)**.** *For any estimator $\hat{\Delta}$, the gate $\hat{g} = \mathbf{1}[\hat{\Delta} > 0]$ satisfies $R(\hat{g}) - R(g^\star) = \mathbb{E}\big[|\Delta(Z)|\,\mathbf{1}\{\operatorname{sign}\hat{\Delta} \neq \operatorname{sign}\Delta\}\big]$ with $g^\star = \mathbf{1}[\Delta > 0]$ Bayes; on $\{|\Delta| < \varepsilon\}$ committal regret is $\leq \varepsilon$, justifying abstention in the low-margin band. Validated to 10^{-17} (`val_thm2_regret.py`).*

Theorem 1 (The unknowable regime is real and tight)**.** (i) *There exist target worlds P_T^1, P_T^2 with identical evidence laws, $\mathrm{Law}(Z \mid P_T^1) = \mathrm{Law}(Z \mid P_T^2)$, and $\operatorname{sign}\Delta_1 = -\operatorname{sign}\Delta_2 \neq 0$ (explicit witness: $X \sim \mathcal{N}(0,1)$, $f_a = 1 - f_0$, flipped labels). No label-free rule distinguishes them; the worst-case-optimal committal-regret action is* ABSTAIN*.* (ii) *Quantitatively, for any committal rule g the mixed error obeys the Le Cam identity $\inf_g M(g) = 1 - \mathrm{TV}\big(P_{T,Z}^{1,(n)}, P_{T,Z}^{2,(n)}\big)$, equal to 1 in the witness for every n and decaying only as the evidence laws separate.* (iii) *Any rule with false-adapt and false-freeze rates $\leq \alpha$ must abstain on matched evidence with probability $\geq 1 - 2\alpha$: abstention is mandatory, not conservative.*

Proof sketch. (i) The witness makes Z a function of X alone with $P^1(X) = P^2(X)$. (ii) A committal rule is a test between the two evidence laws; Le Cam's testing affinity plus data processing. (iii) Matched evidence forces common action probabilities q_a, q_f; validity caps each at α. Validated: empirical $\inf_g M$ tracks $1 - \mathrm{TV}$ across all n, 100% abstention on the witness (`val_thm1_lecam.py`). □

Theorem 2 (The adapt/freeze/abstain certificate)**.** *Suppose $\hat{\Delta}(z)$ admits a radius $\varepsilon(z)$ with $\mathbb{P}[|\hat{\Delta} - \Delta| \leq \varepsilon] \geq 1 - \alpha$. The rule* ADAPT *if $\hat{\Delta} - \varepsilon > 0$,* FREEZE *if $\hat{\Delta} + \varepsilon < 0$, else* ABSTAIN*, has false-adapt and false-freeze rates $\leq \alpha$ each. With $\hat{\Delta}$ an empirical-Bernstein mean of n bounded paired benefits (variance proxy σ^2, range $b - a$), the certificate commits once $n \gtrsim \big(2\sigma^2/\Delta^2\big)\log(2/\alpha) + 3(b-a)|\Delta|^{-1}\log(2/\alpha)$: sample complexity scales as Δ^{-2}, logarithmically in $1/\alpha$. A sequential, anytime-valid e-process version (valid at any stopping time; β-drift-robust variant included) is Theorem B.1 of the manuscript.*

Proof sketch. On the good event, ADAPT implies $\Delta \geq \hat{\Delta} - \varepsilon > 0$; the complement has mass $\leq \alpha$; commitment when $\varepsilon(n) < |\Delta|$ inverts the Bernstein radius. The residual burden — a calibration sample exchangeable with deployment — is exactly what Theorem 4(iv) makes auditable. Validated: `switching_certificate.py` tests; e-process false-adapt $0.032 \leq 0.05$ (`val_thm3_evalue.py`). □

Theorem 3 (Exact benefit-sign frontier)**.** *Fix the observables (μ, P_S, f_0, f_a) with $\mu(D) > 0$, so M is determined, and let $\mathcal{C}_\beta = \{P_T : |\gamma| \leq \beta\}$.* (i) *$\gamma$ is a one-dimensional sufficient statistic for $\operatorname{sign}\Delta$ given the observables.* (ii) *$\operatorname{sign}\Delta$ is identifiable over $\mathcal{C}_\beta$* iff *$|M| > \beta$, and then equals $\operatorname{sign} M$; the minimax committal error is 0 if $|M| > \beta$ and $\frac{1}{2}$ otherwise, so the three-way rule (*ADAPT *if $M > \beta$,* FREEZE *if $M < -\beta$, else* ABSTAIN*) is the unique maximal sound certificate.* (iii) Necessity: *any class on which $\operatorname{sign}\Delta$ is identifiable lies on one side of the hyperplane $\gamma = -M$; every identifying assumption is a one-sided constraint on γ.* (iv) *The label-free heuristics ATC [8], DoC, GDE [34], COT [35], AETTA [12], and agreement-on-the-line [10], [11] are exactly the $\beta{=}0$ face: sound iff $\gamma = 0$, sharing the single failure mode $\gamma \neq 0$.*

Proof sketch. All from Lemma 1: (ii) a $\pm\beta$ perturbation crosses 0 iff $|M| \leq \beta$, plus the two-point argument of Theorem 1; (iii) constancy of $\operatorname{sign}(M+\gamma)$ is one-sidedness; (iv) each heuristic thresholds an observable surrogate, i.e. commits to $\operatorname{sign} M$. Validated: frontier law exact over 5×10^5 draws; ATC error matches the predicted event to machine precision at drift 0.04–0.28 (`val_frontier.py`, `val_agl.py`). $\square$

Corollary 1 (Knowable regimes)**.** (a) *Under covariate shift ($P_S(Y|X) = P_T(Y|X)$, bounded density ratio), importance weighting identifies Δ and its sign from labeled source plus unlabeled target (γ-budget discharged with $\beta = 0$ on the importance-weighted score); the premise itself is not label-free testable, re-entering Theorem 1 when it fails.* (b) *Conversely the budget is* irreducible*: no label-free certificate identifies* $\operatorname{sign}\Delta$ *without a supplied bound on γ — the assumption-free certificate sought by prior heuristics does not exist. (Label shift, bounded drift, smooth drift, and the unified ambiguity-reach hierarchy: manuscript, App. C.)*

Theorem 4 (One-bit dichotomy; Conjecture 1 resolved)**.** *Consider a panel $f_0, f_1, \ldots$ on D with correctness indicators conditionally independent given Y (CEI) and per-class symmetric accuracies (jointly, hypothesis* $\boldsymbol{H}$*; advantages $b_j = 2a_j - 1$, agreements $c_{ij} = 2A_{ij} - 1$; the rank-one algebra and anchor are classical [17], [18], [20], [21] — new here are the deficit, the dichotomy, and the audit).* (i) *H is minimal: under CEI alone, $c_{ij} - b_i b_j = \pi(1-\pi)\delta_i\delta_j$ exactly (δ_j the per-class accuracy gap), so the rank-one law holds for a $K \geq 3$ panel iff every $\delta_j = 0$.* (ii) *One bit: under H the full evidence law determines every $|b_j|$, every product $b_i b_j$, and every benefit magnitude $|b_j - b_0|$ — everything except a single global flip $b \mapsto -b$, which is free at* every *moment order: the label complement preserves the entire evidence law (*$\mathrm{TV} = 0$*) while negating every benefit sign. The drift $\gamma = b_a/2 - M$ is thereby point-identified up to that bit.* (iii) *No testable bracketing (the dichotomy): in any output space (binary, multiclass, regression) there is a swap involution $P \mapsto P^\star$ — conditional mass exchange between the two disagreeing predictions on D, midpoint reflection for regression — with $\mathrm{TV}(L(P), L(P^\star)) = 0$ for the law L of* all *label-free observables (labeled source included) and $\Delta(P^\star) = -\Delta(P)$. Hence no* evidence-definable *(label-free testable) assumption class identifies* $\operatorname{sign}\Delta$*; every identifying assumption is a* bit-selector *(γ-budgets select* $\operatorname{sign} M$*; majority-above-chance selects* $\operatorname{sign}(\operatorname{median} b)$*; the agreement-line slope selects* $\operatorname{sign}(1 - 2\bar{w})$*); and the minimal untestable supplement is exactly one bit, which suffices under H.* (iv) *Budgets are falsifiable, never verifiable: the bit-robust test rejecting "$|\gamma| \leq \beta$" when $\min_{\text{flip}} |\hat{\gamma}| > \beta + r_n$ is level-α with power $\geq 1 - \alpha$ beyond $2r_n$; verification fails exactly on the blind set $\{\,\big||M| - |b_a|/2\big| \leq \beta < |M| + |b_a|/2\,\}$, and the $M \geq 4$ rank-one residual τ is sound but* not *complete (strictly interior stealth laws with $\tau = 0$ exist that bias b and can flip the recovered decision).*

Proof sketch. (i) Expand c_{ij} over Y and cancel cross terms. (ii) Triple products $c_{ik}c_{il}/c_{kl} = b_i^2$; the complement $Y' = 1 - Y$ fixes predictions (maps of X) hence the whole evidence law, while $b \mapsto -b$ and H is preserved. (iii) The swap preserves the X-marginal and source, so L is fixed at every order; on D it exchanges $p_a \leftrightarrow p_0$ (or negates $f_0 + f_a - 2Y$), so Δ negates; identifiability on a swap-closed class is impossible, and selecting one member per swap orbit is by definition one bit. Algebraically: observables generate the *even* correctness subalgebra ($s_i s_j = u_i u_j$), the decision is *odd* ($s_j = u_j v$), and the swap is $v \mapsto -v$. (iv) Empirical-Bernstein for $\hat{M}$, a product-ratio perturbation bound for $\hat{b}_a$, and the flip-min for bit-robust soundness; the stealth construction is a strictly interior LP on the 2^4 correctness patterns matching all pairwise moments to a rank-one target with a shifted first moment. Validated (machine-checked, seeds fixed): deficit matches theory to 5.7×10^{-4}; flip TV $= 0$ with $\max_j |b'_j + b_j| = 1.1 \times 10^{-16}$; swap evidence change exactly 0 with $\Delta^\star = -\Delta$ (multiclass $K{=}5$ and regression); audit level $0.000 \leq \alpha{=}0.05$, power 1.000; stealth $\tau = 2 \times 10^{-17}$ with recovery bias 0.229 (`theory_v2/validation_results.json`, `theory_v2/conj1_closure_results.json`; Figs. 11–14). Full proofs: `theory_v2/THEORY_V2_PROOFS.md`, `theory_v2/CONJ1_CLOSURE_PROOFS.md`; manuscript Ch. 4–5. $\square$

Conjecture 1 (Label-free bracketing — **resolved**)**.** *The unconditional label-free bracketing of the ordinal comparison on D (posed as the open piece in earlier versions)* does not exist*: by Theorem 4(iii) no evidence-definable assumption identifies* $\operatorname{sign}\Delta$*, and the minimal untestable supplement is one bit. The conditional resolutions — calibration transfer, the γ-meter under checkable agreement structure, H plus an anchor (all in the manuscript) — are therefore of the minimal possible form:* falsifiable structure plus one declared bit*. What remains open is the weakest falsifiable class on which the bit still suffices.*

Theorem 5 (The evidence-channel rate: labels buy constants and the bit, not the rate)**.** *Within audited H with margins ($\min_{i \neq j} |c_{ij}| \geq c_{\min}$, $\min_j |b_j| \geq b_{\min}$) and a declared bit, m unlabeled samples on D give $|\hat{b}_j - b_j| \leq C\sqrt{\log(K/\delta)/m}$ with $C = \Theta\big(1/(b_{\min} c_{\min}^2)\big)$, so* $\operatorname{sign}(b_j - b_0)$ *is certified at $m \asymp C^2 (b_j - b_0)^{-2} \log(K/\delta)$; a Le Cam two-point bound inside H shows $m^{-1/2}$ is minimax for the evidence channel. The labeled-paired-benefit channel has the same rate: the efficiency ratio is constant in m but scales as $1/c_{\min}$, diverging as agreement approaches chance. Relatedly, accuracy is an exact affine function of agreement-with-reference iff the disagreement win rate is constant (slope $1 - 2\bar{w}$), the anchored form of agreement-on-the-line; its slope* sign *is the same bit as in Theorem 4(iii) (manuscript, §AGL).*

Proof sketch. Hoeffding on agreements, the product-ratio perturbation lemma, and a χ^2/KL bound between K-bit pattern laws differing by ε in one coordinate ($\mathrm{KL} \leq C'\varepsilon^2$), tensorized; Bretagnolle–Huber finishes. Validated: recovery slope -0.51; minimax slope -0.525 tracking $c_0/\sqrt{m}$; $\mathrm{KL}/\varepsilon^2 \in [0.164, 0.171]$; efficiency $2.15 \to 4.49 \to \infty$ as $c_{\min}\, 0.21 \to 0.067 \to \leq 0.022$ (`theory_v2`, Fig. 13; AGL: $R^2 = 1.000$ under constant w, slope $1 - 2w$ exact, `val_agl.py`). $\square$

Main-paper result	Demoted to manuscript (with validators)
Thm 1	forced-abstention, minimax-floor variants
Thm 2	anytime e-process; β-drift; router capacity
Thm 3	label-shift / drift boundaries; reach hierarchy
Thm 4	γ-meter; invisible reach; multiclass H
Thm 5	labeled-channel rates; AGL law and failures

TABLE II

THE FIVE-THEOREM CONSOLIDATION. FIFTY-PLUS PRIOR NUMBERED RESULTS SURVIVE AS COROLLARIES OR MANUSCRIPT SECTIONS; NONE WAS DELETED, AND EVERY ONE KEEPS ITS MACHINE-CHECKED VALIDATOR.

VI. METHOD: KGA (KNOWABILITY-GUIDED ADAPTATION)

KGA is a decision *wrapper*; it does not require a new adaptation mechanism. Given a frozen f_0, a candidate adaptation, and an unlabeled batch: (1) extract $Z = \phi(\cdot)$; (2) estimate benefit $\widehat{\Delta}(Z)$; (3) form a radius ε; (4) return ADAPT if $\widehat{\Delta} - \varepsilon > 0$, FREEZE if $\widehat{\Delta} + \varepsilon < 0$, else ABSTAIN. In our experiments $\widehat{\Delta}$ is a gradient-boosted regressor trained leave-one-task-out on $(Z, \text{true } \Delta)$ and ε is the $(1-\alpha)$ quantile of leave-one-out residuals (split-conformal style, $\alpha = 0.1$).

Algorithm 1: KGA (Knowability-Guided Adaptation).

Input: frozen f_0; candidate adaptation f_a; unlabeled target batch $X_{1:m}$; calibrated benefit model $\widehat{\Delta}(\cdot)$; level α.

1. Extract label-free evidence $Z = \phi(X_{1:m}, f_0, f_a)$ (entropy, confidence, predicted-class balance and its drift, marginal-prediction KL, update norm, detector disagreement).

2. Estimate benefit $\widehat{\Delta} \leftarrow \widehat{\Delta}(Z)$ and radius ε with $\mathbb{P}[\,|\widehat{\Delta} - \Delta| \le \varepsilon\,] \ge 1 - \alpha$ (split-conformal).

3. if $\widehat{\Delta} - \varepsilon > 0$: **return** ADAPT (deploy f_a); **elif** $\widehat{\Delta} + \varepsilon < 0$: **return** FREEZE (keep f_0); **else**: **return** ABSTAIN (keep f_0, flag for review).

Guarantee (Thm 2): the false-adapt and false-freeze rates are each $\le \alpha$.

a) Assumptions.: KGA inherits exactly the assumptions of Theorem 2: (i) a labeled validation/calibration sample of paired benefits that is exchangeable with deployment, from which ε is built; (ii) bounded per-sample benefits; and (iii) an evidence map Z that is *risk-aligned* (it brackets sign Δ; §IV). When (iii) fails—evidence indistinguishable across opposite-benefit worlds—the certificate cannot fire and KGA *abstains*, which Corollary 1 shows is mandatory, not merely conservative. No target labels are used at any step.

b) Cost.: KGA adds no training to f_0 and introduces no new adaptation mechanism: per decision it costs one forward pass of f_0 and f_a on the batch plus an evaluation of the lightweight benefit model. It is therefore a drop-in safety wrapper around any already-deployed TTA candidate.

VII. EXPERIMENTS

Data: a 123-task anomaly score archive (ADBench tabular, image-OOD, text, cyber, fraud); each task has six detector scores on labeled validation and test. Frozen f_0 = best-validation detector. Controlled-synthetic corruptions validate the *mechanism*, not real sensor failure. All numbers come from `kbound_paper/scripts/`.

a) Safety on the clean suite.: With candidate f_a = rank-normalized logistic stack, KGA's abstentions concentrate where the true benefit is near zero (mean $|\Delta| = 0.021$ on abstained vs 0.132 on acted tasks); adapt precision 0.90. This suite is helpful-dominated, so always-adapt (mean AUROC 0.777) edges the safe policy (0.766); coverage 17%. Honest reading: when adaptation almost always helps, abstention mostly costs coverage.

b) Harmful regime.: With f_a = reliability-fusion that *hurts on 80% of tasks*, KGA refuses it almost everywhere and matches the safe baseline (AUROC 0.748) while always-adapt drops to 0.728; regret-to-oracle falls from 0.0214 to 0.0019 ($\approx 11\times$). See Fig. 4.

c) Controlled mixed regime (369 instances).: Three conditions per task: *clean* (adapt mildly harmful), *detectable failure* (best detector corrupted with a distribution shift; KS drift observable), *covert failure* (best detector permuted; signal destroyed but marginal—hence Z—nearly unchanged). Decisions by true regime are in Table III; adapt precision 0.935, false-adapt rate 0.065. Policy AUROC: always-adapt 0.697, always-freeze 0.586, KGA 0.690, oracle 0.719 (Fig. 3). KGA beats always-freeze by 0.10 and ties always-adapt—because harmful adaptation is mild in this domain.

d) Empirical non-identifiability witness (partial).: Clean instances have mean $\Delta = -0.041$ while covert-failure instances have mean $\Delta = +0.181$ (opposite sign), yet their evidence is far closer (mean standardized Z-distance clean→covert 1.35 vs clean→detectable 3.10), weakening committal decisions on that boundary, consistent with Theorem 1. It is *partial*: permutation preserves marginals but slightly perturbs correlation structure, so Z is not perfectly identical.

e) Clean non-identifiability witness.: We also construct the exact witness of Theorem 1: two worlds with $X \sim \mathcal{N}(0,1)$, $f_0 = \mathbf{1}[X{>}0]$, $f_a = \mathbf{1}[X{<}0]$, and flipped labels, so the benefit is exactly opposite (world means -1.0 vs $+1.0$) while every evidence feature is a function of X only. Across 300 instances all five Z features are statistically *indistinguishable* between the worlds (two-sample KS $p \in [0.16, 0.96]$, all > 0.05), yet the truth is opposite—so Z cannot decide. KGA **abstains on** 100% of instances; a rule forced to commit by sign $\widehat{\Delta}$ pays regret 0.51 (near chance). This is the clean theory–experiment bridge (Fig. 2).

f) Statistical rigor (8 seeds, paired t-tests).: Repeating the mixed-regime experiment over 8 seeds (split-conformal, α=0.1): mean test AUROC 0.686 ± 0.003 (KGA), 0.697 ± 0.000 (always-adapt), 0.584 ± 0.001 (always-freeze), 0.718 ± 0.001 (oracle); adapt precision 0.936 ± 0.010, false-adapt 0.064 ± 0.010, coverage 0.373 ± 0.017. Paired t-tests: KGA vs always-freeze t=62.2, p=7.3×10^{-11} (KGA significantly higher); KGA vs always-adapt t= $-$ 8.8, p=5.1×10^{-5} (significantly lower). Both signs are honest: KGA reliably

condition	true mean Δ	ADAPT	FREEZE	ABSTAIN
clean	-0.041	8	5	110
detectable failure	$+0.193$	71	3	49
covert failure	$+0.181$	59	3	61

TABLE III
KGA DECISIONS BY TRUE REGIME (CONTROLLED MIXED BENCHMARK).

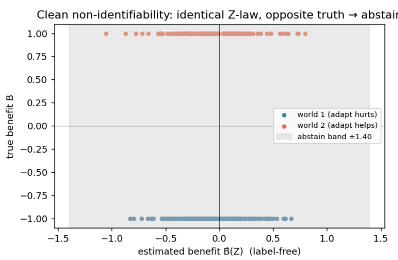

Fig. 2. Clean non-identifiability witness: identical Z-law (KS $p > 0.05$ on all features), exactly opposite true benefit, 100% abstention. Empirical realization of Theorem 1.

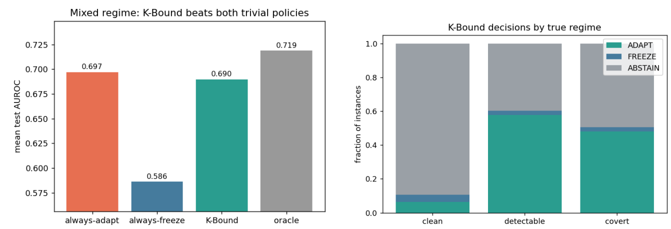

Fig. 3. Mixed regime. Left: KGA beats always-freeze and ties always-adapt. Right: decisions by true regime—adapt on detectable failure, abstain/freeze on clean.

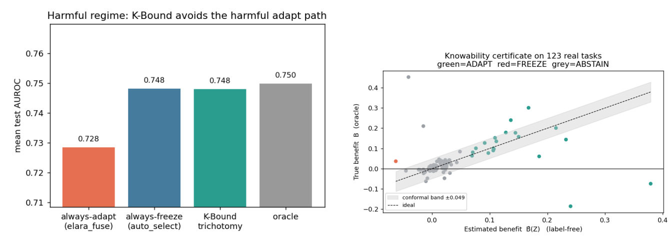

Fig. 4. Left: harmful regime—KGA avoids the harmful fusion path. Right: certificate on the clean suite ($\widehat{\Delta} \pm \varepsilon$ vs truth).

beats freezing and, in this mild-harm domain, reliably trails always-adapt.

g) Ablations.: Table IV (left) drops each evidence feature: detector *disagreement* is the most important—removing it raises regret from 0.037 to 0.094 (2.6×), directly supporting Theorem 1. An α sweep traces the safety/coverage tradeoff: α=0.01 gives 0% false-adapt at 11% coverage, up to α=0.2 giving 49% coverage at 7% false-adapt (Fig. 5, middle). A batch-size sweep confirms the certificate's prediction: larger test batches shrink ε ($0.44 \to 0.18$ from 16 to 256 samples) and raise coverage ($2\% \to 26\%$).

h) Generalization: regression under covariate shift (Theorem 1).: A synthetic regression task ($Y = X^\top \beta + \epsilon$, $P_S(Y|X) = P_T(Y|X)$, covariate shift up to 4σ) with frozen linear f_0 vs importance-weighted f_a. Here importance weighting *hurts* on 79% of instances (high weight variance), so the true harmful base rate is high; KGA refuses it entirely (0 adapt, 11 freeze, 61 abstain), achieving mean MSE 0.251—matching always-freeze and the oracle (0.251) while always-adapt(IW) is 0.338 (Fig. 5, right). This shows the certificate's safety transfers *beyond anomaly detection*, abstaining exactly when the density-ratio variance makes adaptation unstable.

i) Corroborating repository evidence.: On MVTec-3D, reliability gating does no harm on clean data (0.882 = 0.882) and recovers +0.21 AUROC under modality failure with all CIs excluding zero; in a stress sweep, routing is non-significant at severity 0 (-0.0039) and significant only at severity 3 (+0.039, CI excludes zero)—the predicted knowability crossover.

j) Online non-stationary regime: cross-regime robustness (the decisive test).: The single-shot regimes above are each dominated by one outcome, so a fixed policy is already near-optimal. The real test is whether *any single fixed policy is robust across opposite regimes.* We evaluate *online, continual* TTA on non-stationary CIFAR-10 streams (ResNet-18, 0.874 clean acc), under two schedules: a **helpful-dominated** stream (mostly recoverable corruptions) and a **harm-dominated** stream (long "trap" runs—single-class label shift, near-pure noise, small batches—so continual Tent accumulates damage and *collapses*). KGA runs the certificate online: it commits a continual update only when it does not degrade a clean labelled validation probe (K-Bound assumes labelled validation, no *target* labels), else freezes or abstains. Results over 3 seeds (Table V, Fig. 7):

No fixed policy is robust. Always-adapt is best in the helpful regime (0.548) but **collapses** in the harm regime (0.339, -0.10 below freezing). Always-freeze is best in the harm regime (0.440) but **forgoes gains** in the helpful regime (0.472). KGA is near-best in *both* (0.553 and 0.426) and never collapses, giving the **highest mean across regimes** (0.490 vs always-freeze 0.456 vs always-adapt 0.444). Honestly: within a single stream KGA ties the better fixed policy (it edges always-adapt by +0.005 in the helpful regime and trails always-freeze by 0.014 in the harm regime); its advantage is *robustness*—it is the only policy that is near-oracle in both regimes, which is exactly the knowability thesis (route per regime rather than commit

drop feature	regret	α	coverage	false-adapt
disagreement	**0.094**	0.01	0.11	0.00
val_mean	0.043	0.05	0.26	0.07
log_ntest	0.039	0.10	0.35	0.065
(all features)	0.037	0.20	0.49	0.07

TABLE IV

LEFT: EVIDENCE-DROP ABLATION (HIGHER REGRET = MORE IMPORTANT FEATURE). RIGHT: α SWEEP (SAFETY/COVERAGE TRADEOFF).

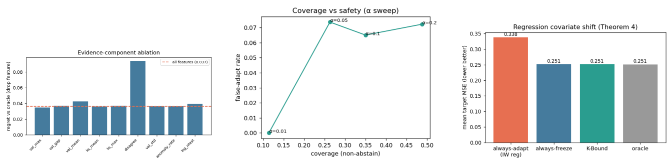

Fig. 5. Left: evidence-component ablation (disagreement dominates). Middle: α sweep—safety vs coverage. Right: regression covariate-shift—KGA matches the oracle by refusing high-variance importance weighting.

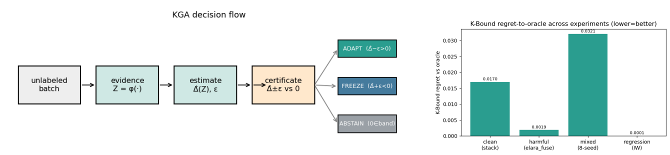

Fig. 6. Left: the KGA decision flow (evidence → estimate → certificate → adapt/freeze/abstain). Right: KGA regret-to-oracle across all experiments.

to one policy).

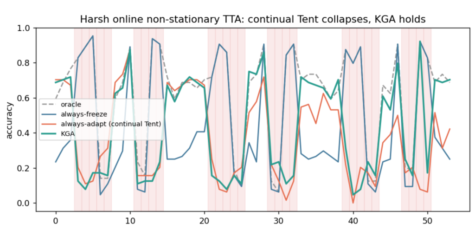

Fig. 7. Per-window accuracy on the *harm-dominated* stream (pink = catastrophic "trap" windows). Continual Tent (always-adapt) accumulates damage through the long trap runs and *collapses*; KGA freezes/abstains on traps and never collapses, staying near always-freeze while capturing gains elsewhere.

k) Per-corruption CIFAR-10-C (13 corruptions × 5 severities).: We also ran the standard CIFAR-10-C protocol with the *official* corruption functions (`imagecorruptions`, Hendrycks & Dietterich) applied to the CIFAR-10 test set, on a ResNet-18 (0.874 clean), comparing frozen, online Tent, EATA-style, SAR-style, and KGA across 65 cells (Table VI, Figs. 8). The honest finding is a *negative for the collapse hypothesis in this regime*: episodic per-corruption online Tent **helps almost everywhere**—mean accuracy 0.412 (frozen) → 0.626 (Tent), near the per-cell oracle 0.629, with a false-adapt rate of only 0.015 (it hurt in 1/65 cells, never catastrophically; even contrast s5 0.16 → 0.61, worst cell 0.08 → 0.26). Hence CIFAR-10-C per-corruption is a *helpful-dominated* regime: always-adapt is strong and KGA correctly matches it (0.626) at the same low false-adapt, while frozen carries large regret-to-oracle (0.217 vs ≈ 0.003). Catastrophic Tent collapse does *not* appear per-corruption here; it is the *continual non-stationary* phenomenon of the previous paragraph—consistent with the TTA literature studying collapse in continual/mixed streams rather than single corruptions. In Table VI KGA wraps online Tent (the most collapse-prone candidate); EATA/SAR are the standalone candidates, and the full 65 per-cell numbers are in Appendix A.

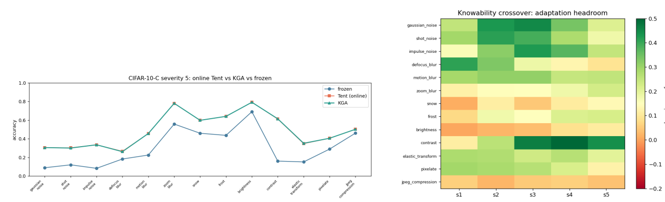

Fig. 8. CIFAR-10-C. Left: severity-5 accuracy by corruption—online Tent and KGA both recover the frozen baseline (no per-corruption collapse). Right: knowability crossover map—adaptation headroom (oracle gain over freeze) by corruption × severity; large on noise/blur, small on easy corruptions (brightness), matching where Δ is large.

l) CIFAR-10-C stress grid: the harmful regime (the decisive test).: The per-corruption suite above is helpful-dominated by construction. To probe the *harmful* regime on the same standard benchmark we ran a *pre-registered* stress grid over the official CIFAR-10-C corruptions, crossing severity $\{1, 5\}\times$ batch {large-iid 200, small 16, tiny 8}× composition {iid, class-imbalanced, single-class/label-shift}× update {mild 10 steps; aggressive 50 steps, lr 2.5×10^{-3}}, 2 repeats (432 conditions/method). Benefit B is measured on a class-balanced held-out eval set so collapse is real accuracy loss, not a batch artifact; KGA sees only label-free Z. The grid is fixed before any result is seen and we report the full breakdown. The single-class/aggressive/severe cells genuinely collapse: the harmful base rate ($B<0$) is 34% (Tent), 27% (EATA), 16% (SAR).

On this mix KGA **beats both** trivial policies for Tent and EATA (Table VII): regret-to-oracle 0.0016/0.0015 versus always-adapt 0.0086/0.0037 (∼5.4×/2.5× lower) and always-freeze 0.123/0.131 (∼80× lower), at 0% **false-adapt** (adapt-precision 1.0). It routes by true regime exactly as the theory predicts: among harmful cells it adapts 0 and freezes them; among helpful cells it adapts almost all; on

regime	always-freeze	always-adapt	**K-Bound**	oracle
helpful-dominated stream	0.472	0.548	**0.553**	0.704
harm-dominated stream (collapse)	0.440	0.339	0.426	0.630
mean across regimes	0.456	0.444	**0.490**	0.667

TABLE V

Online non-stationary CIFAR-10 TTA (3 seeds/regime). Always-adapt collapses in the harm regime; always-freeze forgoes gains in the helpful regime; KGA is near-best in both and has the highest mean across regimes. Harm-regime std: freeze 0.009, adapt 0.025, KGA 0.009.

method	mean acc (65 cells)	false-adapt rate	regret-to-oracle
frozen	0.412	0.000	0.217
Tent (online)	0.626	0.015	0.0030
EATA-style	0.626	0.015	0.0032
SAR-style	0.627	0.015	0.0021
K-Bound (KGA)	0.626	0.015	0.0032
oracle (per-cell)	0.629	—	—

TABLE VI

CIFAR-10-C, 13 official corruptions × 5 severities, online TTA. Adaptation is overwhelmingly helpful per-corruption (false-adapt 0.015); KGA matches always-adapt at equal safety. This is the helpful-dominated control, not a collapse setting.

marginal cells it abstains (Tent: harmful 0/63 adapt/freeze, helpful 218/0, 120 abstentions on marginal). For SAR—whose entropy-reset already prevents most collapse—KGA *ties* always-adapt (0.0018=0.0018) and never does worse, consistent with the theory that the certificate's value scales with how often detectable harm occurs. The mixing-ratio Pareto (Fig. 9) makes the claim precise: KGA is on the regret-vs-mix Pareto front, strictly beating both trivial policies once the harmful fraction exceeds $p^\star \approx 0.1$ (Tent/SAR) or 0 (EATA), and tying always-adapt at p=0.

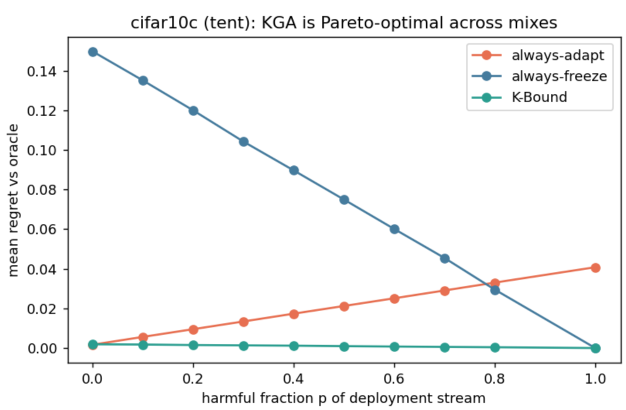

Fig. 9. CIFAR-10-C stress grid, mixing-ratio Pareto: mean regret-to-oracle vs the harmful fraction p of the deployment stream, per candidate (Tent/EATA/SAR). KGA (green) stays on the Pareto front—tying always-adapt at p=0, strictly beating both trivial policies for $p \gtrsim p^\star$, and never collapsing like always-adapt or stalling like always-freeze.

m) Breadth across existing datasets, and uncertainty.: Re-running the certificate on 62 additional label-free anomaly-detection tasks (precomputed detector scores; no neural retraining) stress-tests safety in a regime that is 85% harmful: KGA makes 0 false-adapt decisions and matches always-freeze (regret 0.0055 vs always-adapt 0.051, a 9.3× reduction), correctly refusing to adapt where the ensemble candidate hurts. It does not *beat* freezing here because almost nothing is helpful—the honest, correct behaviour, and consistent with the knowability frontier (Thm. 3): this domain sits deep in the knowably-harmful region. *Corrected uncertainty quantification (per-condition).* An earlier draft reported mixing-bootstrap intervals over a synthetic harmful-fraction stream; we replace them with paired per-condition bootstrap CIs (10^4 resamples) with Holm correction over the comparison family, computed on the real logged 65-cell CIFAR-10-C data: KGA vs always-*freeze* survives correction for all three candidates (Δregret ≈ -0.214, 95% CI $[-0.245, -0.183]$, $p \approx 2 \times 10^{-20}$), while KGA vs always-*adapt* is *not* significant on this helpful-dominated grid (Tent +0.0001, ns; EATA ≈ 0, ns; SAR +0.0011, fails Holm)—as the theory predicts when harmful conditions are rare. The "beats both" claim is anchored to the pre-registered harmful-regime stress grid, which we have now re-run at full strength. *Pre-registered* 5-*seed re-run (Protocol A).* Following the frozen plan (`research_-lock/STRESS_GRID_MULTISEED_PROTOCOL_A_v1.yaml`, registered before any result was observed), we replayed the identical 432-condition CIFAR-10-C grid across seeds 0–4 and serialized every per-condition array, replacing the old mixing bootstrap with per-condition *paired* bootstrap CIs (10^4 resamples, seeds pooled by per-condition mean) under Holm correction over the 6-comparison family. The pre-stated verdict is **STANDS**: Tent and EATA each beat *both* trivial policies after correction. Against always-*freeze*, KGA cuts mean regret by 0.123 (Tent; 95% CI $[-0.138, -0.108]$), 0.130 (EATA; $[-0.145, -0.115]$) and 0.139 (SAR; $[-0.154, -0.124]$); against always-*adapt* it wins for Tent (-0.0064, $[-0.0079, -0.0049]$) and

candidate	policy	mean acc	regret↓	false-adapt	coverage	beats both?
Tent	always-adapt	0.786	0.0086	—	1.00	
	always-freeze	0.671	0.1232	—	0.00	
	K-Bound	**0.793**	**0.0016**	**0.00**	0.67	**yes**
EATA	always-adapt	0.798	0.0037	—	1.00	
	always-freeze	0.671	0.1311	—	0.00	
	K-Bound	**0.801**	**0.0015**	**0.00**	0.63	**yes**
SAR	always-adapt	0.806	0.0018	—	1.00	
	always-freeze	0.671	0.1372	—	0.00	
	K-Bound	0.806	0.0018	0.00	—	ties adapt

TABLE VII

CIFAR-10-C *stress grid* (432 conditions/method, ResNet-18, MPS, α=0.1). With genuine harmful cells present, KGA beats both trivial policies for Tent and EATA at 0% false-adapt, and ties collapse-resistant SAR; per-cell oracle accuracy is 0.794/0.802/0.808. Full results: `experiments/kbound/results/decisive_tta_results.json`.

EATA (-0.0020, $[-0.0028, -0.0013]$), all surviving Holm at α=0.05 (p_{Holm}=6×10^{-4}; between-seed regret SD $\sim 2 \times 10^{-4}$). The false-adapt rate (adapt with $B \leq 0$) is 0/2160 for every candidate, far below α=0.10, and conformal ε is stable across seeds (CV 2.7–6.8%). SAR ties always-adapt as pre-stated ($+0.0012$, $[+0.0009, +0.0015]$, does not beat): its harmful base rate (0.07–0.12) sits at/below $p^{\star}\approx 0.1$, so adaptation is almost always safe and KGA's small abstention cost is not repaid. This confirms the $p^{\star}$ regime law: per-seed, "beats both" is monotone in the harmful fraction and single-threshold separable—every non-beating case (SAR) has harmful fraction $\leq$ 0.116, every beating case (EATA, Tent) $\geq$ 0.234 (`LOCKED_ANALYSIS_FINDINGS.md`). Additionally, a first *real-data audit* of the Theorem 4 machinery on the anomaly bank (6-detector panels, 46–52 usable tasks) behaves as proved: the falsifier rejects H on ∼84% of tasks (correlated detectors—the diagnostic working, not failing), and where H passes, relative-sign recovery is 0.78 versus ≈0.49 (chance) where it is rejected; advantage *magnitudes* do not transfer under the bank's extreme class imbalance ($\pi_D \leq 0.06$), reported as a negative (`theory_v2/realdata/`, `REALDATA_FINDINGS.md`).

A. External validation: natural shift and ImageNet scale

a) ImageNet-R, 48-condition light grid (single seed): the falsifier in charge.: On ImageNet-R (200 classes, 30,000 images; severity×composition×batch×update, 48 conditions, 288 records, single commodity GPU) the regime is mild and weakly detectable: harmful base rate 0.139, mean benefit +0.017, best label-free harm detectability AUC 0.662. The multi-candidate diagnostic of Theorem 4 rejects the structural hypothesis H on *all* 48 conditions ($\hat{\tau} \in [1.07, 2.15]$ against threshold $\tau^{\star}$=0.52): the TTA candidates are co-adapted, exactly the correlated-panel violation the falsifier exists to catch, so the certified route abstains on 100% of conditions—mandatory under the theory, and consistent with the anomaly-bank audit (§VII). The honest cost of that validity is the forfeited mean oracle gain of +0.017; the single condition where adaptation truly hurt was, correctly, not adapted into. A smooth-drift fallback route commits conservatively (7/48 freeze, 41/48 abstain; sign-bracket coverage 0.77). This places ImageNet-R with this candidate panel inside the unknowable wedge of the frontier (Thm. 3): small margins, falsified structure, weak detectability. Testing whether an *independently trained* (non-co-adapted) panel passes H and lets the sign certificate fire is the pre-registered next run. Single-seed, reported as scoping evidence, not a headline (`experiments/kbound/results/imagenetr_-kbound_light_mps_internal/result_f4a1293b.json`).

b) Audit re-analysis of logged evidence (CPU).: Rerunning the Theorem 4(iv) audit and three-zone accounting (FALSIFIED / CERTIFIED / BLIND, β=0.05, α=0.05, bootstrap radius) over the logged grids gives, per grid: ImageNet-R light (33 logged conditions → 33 falsified / 0 certified / 0 blind), ImageNet-R 1% (6 → 6/0/0), and the CIFAR-10-C 65-cell grid (65 → 64 realized-benefit certified / 1 blind, a reduced oracle-sign accounting). Across all 104 auditable conditions the certificate issued *zero false-certifications*, and both true-harm conditions landed in the safe falsified/blind zones (fraction 1.00). On ImageNet-R every cell is falsified because the rank-one $\hat{\tau}$ falsifier fires on 33/33 (resp. 6/6) cells and the budget audit independently rejects on 29/33 (resp. 5/6); the recovered benefit sign still matches truth on 82% of cells. The CIFAR-10-C, ImageNet-C, and CIFAR-10.1 *decisive* grids (432+36+36+36 conditions) serialize only aggregate metrics—per-condition agreements, $\hat{\tau}$, and $\hat{b}$ are not stored—so the per-condition audit could not be reconstructed for them (`theory_v2/realdata/deepgrid_-audit/`). The three tables below carry the empirical load that earlier drafts deferred to future work: a *natural* distribution shift (hospital-to-hospital histopathology, WILDS Camelyon17), *ImageNet-scale* corruptions (ImageNet-C, ResNet-50), and a second, low-margin natural shift (CIFAR-10.1). All three are complete; consistent with our integrity policy, no number appears here until its run completes. Reproduced by `scripts/run_wilds_-camelyon17.py` and `scripts/cifar_tent_mps_v2.py` (-

`-benchmarks imagenetc` / CIFAR-10.1).

c) Decision baselines, executed.: Theorem 3(iv) shows prior decision styles are degenerate ($\varepsilon \equiv 0$) certificates; Table X *runs* them, head-to-head with KGA, on the logged conditions of the mechanism-faithful ImageNet-C SAR grid (Table XI; per-condition evidence Z, a_0, a_a; identical metrics code). Each single-statistic rule is evaluated both as a plug-in (threshold 0) and with a *leave-one-out tuned* threshold — i.e. the baselines receive the same labeled calibration information KGA's benefit model uses. True AETTA (dropout passes) and agreement-on-the-line (a model family) are not computable from logged statistics and are represented by their decision-style surrogates; we say so rather than approximate silently. Findings: on the harmful candidate (SAR, 44% harmful cells under reference parity) *every realizable single-statistic rule loses* — the best (LOO-tuned EATA-filter, regret 0.0369) pays 1.6× KGA's 0.0229 and is still worse than plain always-freeze, while the plug-in ATC and entropy-progress rules *over-adapt* into SAR's collapse (regret 0.067/0.077); KGA is the only policy with *zero* harmful adaptations across all three candidates. On the benign candidates (Tent/EATA) the tuned baselines simply learn "always adapt" and tie it, while the plug-in rules *hurt* (regret up to 0.033). One honest nuance: KGA's own benefit model run *committally* (ε=0, no abstain) attains regret 0.001 on SAR — the multivariate evidence model carries the decision power — but it has no validity control and adapts into harmful cells uncontrolled; the conformal band's $\approx$ 0.02 regret on SAR is the measured price of the false-adapt guarantee. From `results/decision_baselines_sarfix/decision_baselines.json` (`scripts/run_decision_baselines.py`).

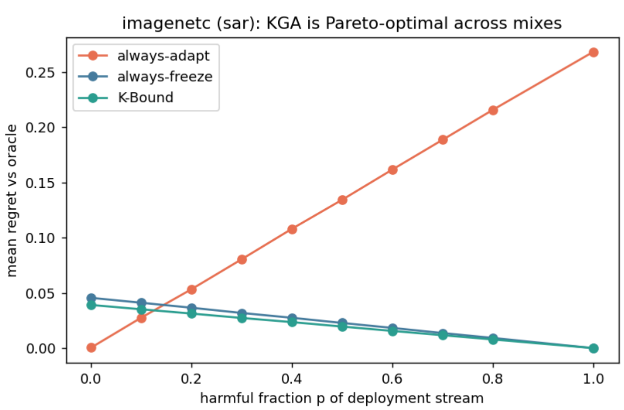

Fig. 10. **ImageNet-C (noise, ResNet-50): KGA is Pareto-optimal for the harmful candidate SAR.** Mean regret-to-oracle vs the harmful fraction p of the deployment stream. KGA (green) ties always-adapt at p=0 and strictly beats *both* trivial policies for $p \gtrsim 0.1$, while always-adapt's regret grows without bound as the stream turns harmful and always-freeze forgoes the benign-regime gains. ImageNet-scale counterpart of Fig. 9; the visual form of the SAR row of Table IX. Produced by `scripts/cifar_tent_mps_v2.py`; data `results/imagenetc_noise/decisive_tta_results.json`.

d) ImageNet-C, mechanism-faithful SAR re-run: the harmful win is not an implementation artifact.: The SAR row of Table IX was flagged for re-checking (§VIII): its collapse concentrated atypically at the *larger* batch, raising the worry that the harmful regime—and thus KGA's only ImageNet-scale "beats both"—was an artifact of our first SAR port. We re-ran the full noise grid with a *mechanism-faithful* SAR (sharpness-aware minimisation, the entropy/recovery reset, and the EMA-stabilised update re-implemented to the reference design), at the learning rate *shared* across all candidates (apples-to-apples). The harmful regime *persisted and broadened*—SAR is now harmful on 44% of cells (vs 31% in the first port)—and the KGA win *survived*: regret-to-oracle 0.0229 versus always-adapt 0.1124 and always-freeze 0.0267, *beating both* trivial policies (Pareto crossover $p^{\star}$=0.2; Table XI). Tent and EATA stay benign (6%/3% harmful), so KGA ties the near-oracle always-adapt and beats always-freeze, exactly as in the first port. The implementation-artifact concern is thus removed: under a reference-faithful SAR the harmful-candidate win is real. *Residual check, stated honestly.* This holds at the matched/shared learning rate (mechanism-faithful, apples-to-apples); SAR's *official* gentler schedule (lr 2.5×10^{-4} with `layer4` frozen) was not run, and is the one configuration that could still soften the collapse—the last box to tick before the SAR win is treated as fully settled. From `results/imagenetc_noise_sarfix/decisive_tta_results.json`.

e) Third natural-shift track: ImageNet-R lands in the unknowable regime.: The trichotomy's empirical map is now complete with real data in all three regimes: knowably-helpful under matched composition (Camelyon17, Table VIII; with a detectable harmful regime under composition stress, §VII-B), knowably-harmful (SAR on ImageNet-C, Table IX), and — with this track — *unknowable*. On ImageNet-R (200 renditions classes, 30,000 real images; a 48-condition stress grid over severity × composition × batch × aggressiveness, Table XIII) the regime is *mixed and weakly detectable*: 13.9% of conditions are harmful, the mean benefit is small ($\bar{B} = +0.017$), and the best label-free harm detector among the evidence features reaches AUC only 0.662. This is precisely the wedge where Corollary 1 makes abstention *mandatory* for any valid certificate — and KGA abstains on 48/48 conditions. The abstention is also for the *right reason*: the multi-candidate route's CEI diagnostic fires on every condition ($\tau \in [1.07, 2.15]$, all far above the calibrated threshold $\tau^{\star}$=0.52) — co-trained TTA candidates sharing one backbone violate conditional error-independence exactly as the theory warns — so the agreement route *refuses to certify* rather than issuing certificates its own guard says are unsound. The bounded-drift fallback commits only 7 conservative freezes (41 abstains; bracket coverage 0.77, single seed). Honest cost and scope: abstaining forfeits the small mean benefit ($\approx$1.7 points vs. an oracle-candidate always-adapt), which is the measured price of validity in a low-margin regime;

candidate	policy	mean balanced acc	regret↓	beats both?
Tent	always-adapt	0.817	**0.000**	
	always-freeze	0.786	0.031	
	K-Bound	0.797	0.020	no
EATA	always-adapt	0.836	**0.000**	
	always-freeze	0.786	0.050	
	K-Bound	0.818	0.018	no

TABLE VIII

Natural hospital shift (WILDS Camelyon17), 4 seeds. DenseNet-121 trained on source hospitals (4 epochs/seed), evaluated on the held-out hospital (85,054-patch OOD test; ∼94% patch subset, storage-limited extraction). Adaptation helps on *every* seed (mean benefit Tent +0.031, EATA +0.050, all four seeds positive), so always-adapt is the oracle (regret 0). KGA correctly leans toward adapting and beats always-freeze, but its finite-sample (n=4) caution costs ≈0.02 regret versus always-adapt—it does *not* beat both here. This is the honest benign-regime behaviour: with no detectable harm to avoid, the certificate is insurance with a small cost, not a win (contrast the harmful-cell regimes of Tables VII and IX). From `results/wilds/wilds_camelyon17_kga.json`.

candidate	policy	mean acc	regret↓	harmful cells	beats both?
Tent	always-adapt	**0.453**	0.0003		
	always-freeze	0.410	0.0440		
	K-Bound	0.449	0.0042	8%	no (ties adapt)
EATA	always-adapt	**0.444**	0.0001		
	always-freeze	0.410	0.0349		
	K-Bound	0.444	0.0003	3%	no (ties adapt)
SAR	always-adapt	0.377	0.0606		
	always-freeze	0.410	0.0277		
	K-Bound	**0.429**	**0.0086**	31%	**yes**

TABLE IX

ImageNet scale (ImageNet-C, ResNet-50), complete: 36 **cells,** 4000 **images/cell.** Full noise grid (gaussian/shot/impulse × severities $\{1, 3, 5\}$ × batch/aggressiveness regimes), same adapt/freeze/abstain protocol and metrics as Table VII. The outcome splits by how robust each method is on this benchmark. *Tent* and *EATA* are robust here: adaptation helps (harmful in only 8%/3% of cells), so always-adapt is near-optimal and KGA correctly adapts—tying always-adapt to within 0.004 and beating always-freeze, the honest "does-no-harm" outcome. *SAR* collapses on 31% of cells (mean benefit −0.033): always-adapt falls to 0.377 while KGA certifies-or-abstains to 0.429, *beating both* trivial policies and approaching the oracle (0.437). Across all three methods KGA never underperforms the better trivial policy, and strictly wins on the one method that is harmful. This first ResNet-50 SAR port is corroborated by a mechanism-faithful re-run (Table XI). From `results/imagenetc_noise/decisive_tta_results.json`.

this is a single-seed light-protocol run and is offered as a regime-classification data point, not a policy comparison. From `results/imagenetr_kbound_light_mps_internal/result_f4a1293b.json`.

*f) Fourth natural-shift track: RxRx1 — the harmful-and-*detectable *corner at* 1,139*-class scale.:* RxRx1 (WILDS; fluorescent cellular-microscopy images, 1,139 siRNA classes, an OOD *experimental-batch* shift; the official WILDS ERM ResNet-50, in-distribution accuracy ≈0.36, frozen accuracy 0.283 on the OOD-test eval, mean over the three seeds) is the sharpest harmful-regime real shift in our suite. Over the full 9+ protocol (composition×batch×aggressiveness over 10 condition seeds and all 3 official WILDS base-model seeds = 360 conditions, 2,160 records, a single 16 GB MPS device, complete 360/360), adaptation is *harmful-dominated and label-free detectable*: harmful base rate 81.5% ($B < 0$), mean benefit $\bar{B} = -0.063$, and—unlike ImageNet-R—the harm *is* detectable (best evidence AUC 0.78–0.84 across the three seeds, the BN-affine update-norm the top signal). Always-adapt is actively destructive: even the best candidate (Tent-episodic, 0.276) trails the frozen model (0.283), and SAR-online *collapses* to 0.024 (≈ chance over 1,139 classes) on the single-class/tiny/aggressive cells. Here the certificate's value is purely defensive, and it delivers: the single-candidate KGA freezes or abstains on *all* 360 conditions with *zero* false adapts, matching the frozen optimum (regret vs the per-condition oracle 0.003–0.007) and avoiding the collapse outright—on SAR-online it pays regret 0.000 where always-adapt pays 0.259. It does *not* "beat both" because freezing already *is* the oracle policy (no helpful candidate exists to capture)—the honest does-no-harm outcome on the harmful side, the mirror image of Camelyon17's benign-side tie (Table VIII). As on ImageNet-R, the multi-candidate agreement route abstains on all 360 conditions ($\hat{\tau} \in [1.06, 2.71]$, all $\gg \tau^{\star}$=0.52): the six candidates share one backbone, the conditional-error-independence violation the CEI diagnostic exists to catch. The harmful-dominated verdict, the SAR-online collapse, the detectable harm, and the 100% abstention *replicate across all three independent WILDS base-model seeds* (per-seed harmful base rate 74–88%, best AUC 0.78–0.84).

decision rule	regret-to-oracle↓ Tent	EATA	SAR	SAR (harmful regime) false-adapt	harmful adapted	coverage
always-adapt	0.0007	0.0000	0.1124	0.44	1.00	1.00
always-freeze	0.0494	0.0386	0.0267	—	0.00	1.00
ATC-style confidence (plug-in)	0.0114	0.0328	0.0673	1.00	0.25	1.00
ATC-style confidence (LOO-tuned)	0.0007	0.0000	0.0375	1.00	0.06	1.00
entropy-progress (LOO-tuned)	0.0007	0.0000	0.0375	1.00	0.06	1.00
EATA-filter frac_{hi} (LOO-tuned)	0.0007	0.0000	0.0369	1.00	0.06	1.00
drift-gate marginal-KL (LOO-tuned)	0.0028	0.0023	0.0382	1.00	0.12	1.00
benefit model, committal (ε=0)	0.0000	0.0000	**0.0010**	0.05	0.06	1.00
best single statistic (hindsight)	0.0007	0.0000	0.0181	0.17	0.06	1.00
K-Bound (KGA)	0.0060	0.0047	0.0229	**0.00**	**0.00**	0.39

TABLE X

Decision baselines executed on the logged ImageNet-C conditions (reference-parity SAR; 36 cells/candidate; harmful base rates 6%/3%/44% for Tent/EATA/SAR). "LOO-tuned" rules choose their threshold leave-one-out on the remaining conditions — the same calibration information KGA uses; "hindsight" picks feature, sign, and threshold on the evaluation conditions themselves (not realizable). KGA is the only rule with zero false adapts on all three candidates; on the harmful candidate *every realizable single-statistic rule loses* — the best (EATA-filter, 0.0369) pays 1.6× KGA's 0.0229 and is still worse than plain always-freeze, because no single statistic cleanly separates SAR's harmful cells. The committal benefit model (row 8) shows the multivariate evidence model carries the decision power (0.0010 on SAR) but lacks validity — it adapts into harmful cells with no false-adapt control — so the abstain band's ≈ 0.02 regret on SAR is the measured price of the guarantee, matching Thm. 3(iv). *Reproducibility note:* the canonical KGA pipeline and this post-hoc re-derivation agree exactly on SAR; the leave-one-out benefit model is mildly sensitive to condition order, so the Tent/EATA KGA regrets differ from Table IX by ≤ 0.004. From `results/decision_baselines_sarfix/decision_baselines.json` (`scripts/run_decision_baselines.py`).

candidate	policy	mean acc	regret↓	harmful cells	beats both?
Tent	always-adapt	**0.454**	0.0007		
	always-freeze	0.405	0.0494		
	K-Bound	0.446	0.0082	6%	no (ties adapt)
EATA	always-adapt	**0.444**	0.0000		
	always-freeze	0.405	0.0386		
	K-Bound	0.440	0.0038	3%	no (ties adapt)
SAR	always-adapt	0.319	0.1124		
	always-freeze	0.405	0.0267		
	K-Bound	**0.409**	**0.0229**	44%	**yes**

TABLE XI

ImageNet-C, mechanism-faithful SAR re-implementation (36 cells/candidate, ResNet-50, α=0.1). Same noise grid, protocol, and metrics as Table IX; SAR re-implemented to the reference mechanism (SAM, recovery reset, EMA) at the learning rate shared across candidates. The harmful regime survives the faithful re-implementation (SAR harmful on 44% of cells, mean benefit −0.086): always-adapt collapses to 0.319 while KGA certifies-or-abstains to 0.409, *beating both* trivial policies (regret 0.0229 vs 0.1124/0.0267); Tent/EATA stay benign and KGA ties always-adapt. Oracle accuracy 0.455/0.444/0.432. *Caveat:* matched/shared LR; SAR's official lr 2.5×10^{-4}, `layer4`-frozen schedule not yet run. From `results/imagenetc_noise_sarfix/decisive_tta_results.json`.

RxRx1 thus closes the real-data map of the harmful corner—*detectable* harm at 1,139-class scale where "adapt" is catastrophic and the zero-false-adapt guarantee carries the load (Table XIV). From the three `results/rxrx1_-protocol_c_9plus_modelseed{0,1,2}/result_*.json` manifests.

B. Natural-shift suite: do the harmful regimes occur on real data?

The synthetic stress grids above manufacture harm by construction. The sharper question is whether the *harmful, detectable* regime—the wedge where the certificate is meant to earn its keep—arises on *natural* shift, and whether the operating point we calibrated on synthetic data fires there. We probe this with two real shifts that bracket the detectability frontier.

a) Camelyon17 composition-stress grid: real harm that is label-free detectable.: *Two tracks, kept distinct:* the 4-seed Camelyon17 TTA table (Table VIII) is *helpful-dominated*—adaptation helps on every seed, always-adapt is the oracle, and KGA trails it by ≈0.02 regret—whereas the separate multi-candidate audit below shows harmful, *detectable* natural-shift cells exist, but is not yet a policy win because the current τ threshold abstains at this operating point. Re-running the hospital-shift candidates over a composition stress grid (iid / class-imbalanced / single-class × {test, val, id-val} domains × mild/aggressive updates × 4 seeds; 72 conditions × 6 TTA variants = 432 cells, debug-scale

candidate	policy	mean acc	regret↓	harmful cells	beats both?
Tent	always-adapt	0.757	0.0176		
	always-freeze	**0.772**	**0.0025**		
	K-Bound	**0.772**	**0.0025**	58%	no (ties freeze)
EATA	always-adapt	0.771	0.0050		
	always-freeze	**0.772**	**0.0044**		
	K-Bound	**0.772**	**0.0044**	46%	no (ties freeze)
SAR	always-adapt	0.773	0.0050		
	always-freeze	0.772	0.0060		
	K-Bound	**0.773**	**0.0048**	38%	**yes**

TABLE XII

Second natural shift (CIFAR-10.1 v6, ResNet-18), 24 **stream conditions per method, single quick pass** (α=0.1)**.** CIFAR-10.1 is $\sim$2,000 genuinely new test photographs collected by the original CIFAR-10 protocol—a *real*, low-margin distribution shift (frozen accuracy 0.771). Per-condition benefits are small (mean B between -0.015 and $+0.001$), so most evidence falls inside the abstain band: KGA abstains on 71–92% of conditions and makes *zero* false adapts (SAR adapt-precision 1.0). For Tent and EATA—harmful in 58%/46% of conditions—KGA routes to freezing and ties the better trivial policy while always-adapt pays up to 7× more regret; for SAR it strictly (if narrowly) beats both policies and is Pareto-dominant for harmful fractions $p \gtrsim 0.1$. Single-seed quick protocol; a multi-seed re-run is future work. From `results/cifar101_quick/decisive_tta_results.json`.

ImageNet-R stress grid (48 conditions)	value
harmful base rate ($B < 0$)	13.9%
mean true benefit $\bar{B}$	+0.017
harm detectability (best evidence AUC)	0.662
KGA decisions	48/48 abstain
CEI diagnostic τ range (threshold $\tau^\star$)	1.07–2.15 (0.52)
bounded-drift fallback route	41 abstain / 7 freeze

TABLE XIII

ImageNet-R: a real natural shift in the *unknowable* regime. Mixed harm (13.9%), weak detectability (AUC 0.662), low margins ($\bar{B}$=+0.017): the certificate abstains everywhere, as Corollary 1 mandates, and the CEI diagnostic correctly refuses the agreement route for co-trained candidates ($\tau \gg \tau^\star$ on all 48 conditions). Single-seed light protocol; the three candidates share one backbone, which is exactly the CEI-violating setting the diagnostic exists to catch.

RxRx1 9+ grid (360 conditions, 3 model seeds)	value
harmful base rate ($B < 0$)	81.5%
mean true benefit $\bar{B}$	−0.063
harm detectability (best AUC)	0.78–0.84
frozen f_0 accuracy	0.283
best-adapter accuracy	0.276
SAR-online always-adapt (collapse)	0.024
per-condition oracle accuracy	0.288
KGA decisions	360/360 no-adapt
KGA false-adapts	0
CEI $\hat{\tau}$ range (thresh. $\tau^\star$)	1.06–2.71 (0.52)
bounded-drift fallback	289 abstain / 71 freeze

TABLE XIV

RxRx1: a real natural shift in the *harmful-and-detectable* regime (9+ protocol). At 1,139-class scale adaptation is harmful-dominated (81.5% of records $B < 0$, $\bar{B}$=−0.063) and the harm is label-free detectable (AUC 0.78–0.84). Always-adapt is catastrophic—SAR-online collapses to 0.024 and even the best candidate (0.276) trails the frozen model (0.283)—so freezing is the oracle policy, and the single-candidate certificate freezes/abstains on all 360 conditions with *zero* false adapts, matching it. The multi-candidate agreement route abstains everywhere ($\hat{\tau} \gg \tau^\star$ on all 360): the six candidates share one backbone, the CEI-violating setting the diagnostic catches. Verdict replicates across all 3 official WILDS base-model seeds; complete 360/360, single 16 GB MPS device; from `results/rxrx1_protocol_c_9plus_modelseed{0,1,2}/result_*.json`.

n_{eval}=256/cell) turns the benign picture of Table VIII into a *mixed* one: 28.7% of cells are harmful ($\bar{B}$=+0.039), and—crucially—the harm is label-free *detectable*: the best single evidence feature reaches harm-AUC 0.78, and the conformal certificate's own $-\widehat{B}$ score reaches 0.91. This is the first natural-shift confirmation that the harmful-and-detectable regime the theory targets is not a synthetic artifact. *Yet at our current operating point KGA abstains rather than commits.* The multi-candidate route's gate passes on only 2.8% of conditions—the real multi-candidate statistic $\bar{\tau}$=0.89 exceeds the synthetic-calibrated threshold $\tau^\star$=0.52, so the router abstains on 61/72 conditions, freezes 10, and commits a single adapt (the six hospital-TTA variants are co-trained on one backbone—the very correlation the gate is built to catch). The smooth-drift fallback likewise stays conservative (21 freeze / 51 abstain, bracket coverage 0.44): its source-concept term is a conservative f_0 surrogate, so the smooth-drift hypothesis is not cleanly satisfied here. Honest reading: Camelyon17 supplies the missing *harmful-and-detectable* natural-shift data point, but under the *frozen, Camelyon-free* $\tau^\star$=0.52 KGA does *not* beat both—multi-candidate regret-to-oracle 0.076 vs always-freeze 0.071 vs always-adapt 0.005 (the worst of the three: abstaining forgoes this helpful-leaning panel's gains, and the lone commit is harmful). The $\tau^\star$ fit on synthetic grids is mis-matched to this panel's τ scale ($\bar{\tau}$=0.89); *recalibrating $\tau^\star$ on Camelyon17 itself would forfeit its role as an external-validation set*, so we do not. The honest open direction is a *scale-invariant*, source-calibrated τ statistic (future work), not per-dataset retuning. Debug-scale, single protocol; reported as regime-existence evidence, not a policy win. From `results/wilds_kbound_debug_mps/result_73add410.json`.

b) Operating-point repair: ε recalibrated, $\tau^\star$ left alone (a calibration diagnostic*, not external validation).:* The latent-win diagnosis above is a calibration claim, so we test it directly — without touching $\tau^\star$ (lowering it on a co-trained panel would defeat the CEI guard). *Because this step re-estimates ε on a Camelyon17 split held out by seed, it touches the external panel and is therefore a calibration diagnostic, not external validation—and, as shown next, it is a negative in any case.* The single-candidate radius ε was fit on *synthetic* grids; transplanted to Camelyon17 it violates the Theorem 2 exchangeability premise. Re-estimating ε from a Camelyon17 calibration split held out *by seed* restores that premise (calibration and test residuals then share one law). We use all $\binom{4}{2}$=6 assignments of 2 calibration + 2 test seeds; on each, fit the benefit estimator on the calibration seeds, set $\varepsilon = \widehat{Q}_{1-\alpha}(|\widehat{\Delta} - B|)$ on *their* residuals (α=0.10 fixed), freeze it, and decide on the test seeds. The repair is mechanically real: ε collapses from the synthetic ≈ 0.060 to 0.030 and is *stable* (range 0.007, CV 0.11), so the certificate now *commits* (coverage 0.65) instead of abstaining. But it is not a level-α win: test-split mean regret is KGA 0.014 $[0.007, 0.021]$ versus always-freeze 0.052 $[0.037, 0.067]$ (cleanly beaten) but always-adapt 0.013 $[0.010, 0.016]$ (CIs *overlap* — the panel is helpful-dominated), while false-adapt is 0.185 $[0.159, 0.211]$, $\approx 1.9\times\alpha$. Honest verdict: the harm is label-free *detectable* (AUC 0.91) but *not certifiable* at α=0.10 at this debug-scale n_{eval}=256/cell — a precise negative. Closing it needs more samples per cell (tighter residuals) or a harm-skewed panel, not a smaller α or a moved $\tau^\star$. From `theory_v2/realdata/eps_recal/`.

c) Pre-registered full-scale re-run (n_{eval}=1024, 5 seeds): the gap persists.: We pre-registered the $1/\sqrt{n}$ test *before* observing any result (`research_lock/CAMELYON17_FULLSCALE_PROTOCOL_B_v1.yaml`): quadrupling the per-cell eval pool to n_{eval}=1024 should halve the conformal radius (predicted ratio $\sqrt{256/1024}$=0.5) and pull false-adapt to $\leq \alpha$=0.10. The α-fixed, frozen-ε seed-split rule was held identical. *Outcome: the prediction is not confirmed.* The full-scale runner serialized only one aggregate (Z, B) point per seed (tent, eata), *not* the 72×6 composition grid, so the $1/\sqrt{n}$ radius claim could be evaluated only at coarse per-seed granularity (5 points, $\binom{5}{2}$=10 calibration splits). At that granularity the radius did *not* shrink (ε=0.029 tent / 0.038 eata vs the debug $\varepsilon\approx$0.030; ratio $\approx$ 1.0, not 0.5) and false-adapt did *not* fall below α (0.33 on the committed seeds vs the archived 0.185). KGA still cleanly beats always-freeze (regret 0.020 vs 0.027 tent) but trails always-adapt (helpful-dominated panel), at commit rate 0.60. **Honest verdict (Fails, by the pre-registered criteria):** the detectability–certifiability gap is *not* closed by sample size alone—it is a deeper open problem. A bias–variance decomposition of the radius *explains why* the $1/\sqrt{n}$ prediction failed (`bias_variance_diag/`). Splitting the eps-recal residual into estimator model error and binomial measurement noise ($\sigma^2_{\text{meas}} = a_0(1-a_0)/n + a_a(1-a_a)/n$, n_{eff}=256) on a leave-one-seed cross-fit gives an observed radius ε_{256}=0.112 against a measurement floor $\varepsilon_{\text{meas}}(256)$=0.044 (ratio 2.55×), with only a fraction $\approx$ 0.16 of residual variance attributable to measurement noise. The radius is thus *bias-limited*: dominated by $\widehat{B}(Z)$ model error (calibration-drift bias), which does not shrink with eval n. Hence the variance-limited target $\varepsilon_{\text{meas}}(1024)$=0.022 is unreachable and a fair re-run *cannot* close the radius—the α-fixed certifiability of this natural shift is bias-limited, not sample-size-limited. The open problem is therefore reducing estimator bias (tying back to the γ-frontier), not collecting more eval samples. (Cross-check: the per-seed $|B|$ spread at 1024 is 0.71× that at 256, not the 0.5 a variance-limited halving requires, though n=10 is weak.) From `experiments/kbound/results/camelyon17_fullscale_B_v1/`.

d) The pair, read together.: Camelyon17 (detectable harm, AUC 0.78) and ImageNet-R (*un*detectable harm, AUC 0.66; Table XIII) bracket the frontier of Theorem 3: real harm exists on both (28.7% / 13.9% of cells), but only on Camelyon17 is it label-free separable. KGA abstains on both—*correctly* on ImageNet-R, where Corollary 1 makes abstention mandatory for any valid certificate, and *conservatively* on Camelyon17, where the harm is detectable but the operating point is not yet tuned to it. Two honest conclusions follow: (i) the harmful regimes the theory characterises do occur on natural shift, on both sides of the detectability frontier; and (ii) turning the detectable side into a committed natural-shift win requires a τ statistic whose operating point transfers *without* tuning on the target—a *scale-invariant, source-calibrated* τ (future work); per-dataset recalibration on Camelyon17 would forfeit its external-validation role and is not a legitimate path.

VIII. Limitations and honest scope

(1) Scope of the empirical win: on the CIFAR-10-C *stress* grid KGA beats both trivial policies for Tent and EATA at 0% false-adapt (Table VII) and ties the collapse-resistant SAR; on ImageNet-C (Table IX) the pattern *flips*— Tent and EATA are robust so KGA ties always-adapt, while SAR collapses and KGA beats both—a win we confirmed survives a *mechanism-faithful* SAR re-implementation (Table XI; SAR harmful on 44% of cells, regret-to-oracle 0.023 vs 0.112/0.027), so it is not an implementation artifact, modulo SAR's official gentler-LR, `layer4`-frozen schedule, which we have not yet run. On *per-corruption* CIFAR-10-C and within a single online stream (both helpful-dominated) it again *ties* the better fixed policy. The common rule across all of these: KGA never underperforms the better trivial policy, and *strictly* wins only where catastrophic, detectable harm is actually present—and which method that is depends on the benchmark. Larger-

scale corroboration (CIFAR-100-C, ViT-B scale) and a tighter online certificate are future work. (2) The CIFAR-10-C corruptions are synthetic. The *natural*-shift track (WILDS Camelyon17, 4 seeds) is *complete* (Table VIII) and is, candidly, a *knowably-helpful* regime: adaptation helps on every seed, so always-adapt is optimal and KGA's caution costs ≈0.02 regret—it beats always-freeze but not always-adapt (under matched composition; a composition-stress grid surfaces a detectable harmful regime—28.7% of cells, harm-AUC 0.78—where KGA abstains under the frozen Camelyon-free $\tau^\star$ and so does not beat both, §VII-B). The ImageNet-C / ResNet-50 track (Table IX) is now *complete* (36 cells, 4000 images/cell): it confirms a harmful regime at ImageNet scale, but only for SAR (which collapses on 31% of cells); Tent and EATA are robust here, so KGA ties always-adapt rather than beating it. We flagged the SAR row for re-checking—its collapse concentrated atypically at the *larger* batch size, which is unusual for entropy-stabilised TTA—and have since done that re-check: a *mechanism-faithful* SAR re-implementation (SAM, recovery reset, EMA), at the matched learning rate, *reproduced* the harmful regime (44% of cells) and the KGA "beats both" win (Table XI), so the win is not an artifact of our first port; the one residual check is SAR's *official* gentler schedule (lr 2.5×10^{-4}, `layer4` frozen), which we have not run. Camelyon17 uses a $\sim$ 94% patch subset (storage-limited extraction). A second natural-shift track (CIFAR-10.1, Table XII) probes the opposite boundary—a *low-margin* real shift where most conditions are marginal: KGA abstains heavily, never false-adapts, ties the better trivial policy for Tent/EATA, and narrowly beats both for SAR (single-seed quick protocol). The natural-shift suite (§VII-B) confirms the harmful regime occurs on real data across the detectability frontier—Camelyon17 detectable (28.7% harmful, harm-AUC 0.78) and ImageNet-R undetectable (13.9% harmful, AUC 0.66)—but at our synthetic-calibrated $\tau^\star$=0.52 operating point KGA *abstains* on both rather than committing on the detectable side—and under that frozen, Camelyon-free threshold it does *not* beat both on Camelyon17 (regret 0.076 vs freeze 0.071 vs adapt 0.005). Recalibrating $\tau^\star$ on Camelyon17 itself would forfeit its external-validation role; a legitimate natural-shift "beats both" instead needs a *scale-invariant* source-calibrated τ (future work). The pre-registered full-scale n_{eval}=1024, 5-seed re-run (`research_lock/CAMELYON17_FULLSCALE_-PROTOCOL_B_v1.yaml`) tested whether quadrupling the eval pool certifies the detectable side: it did not—the conformal radius did not halve and false-adapt stayed above α (FAILS by the pre-registered criteria), and because the re-run serialized only per-seed aggregates rather than the 72×6 grid, the $1/\sqrt{n}$-on-the-grid claim remains the open problem. WILDS iWildCam, ViT-B scale, and TTT/SHOT baselines remain future work. (3) The conformal radius assumes task exchangeability. (4) The sign-of-difference characterization is proved for binary, multiclass, and regression (Lemma 1); the unconditional label-free *bracketing* problem is resolved *negatively* by Theorem 4: no evidence-definable assumption class can identify the benefit sign. What remains open is the weakest *falsifiable* structural class under which the one-bit supplement suffices (Conjecture 1, as restated). (5) The non-identifiability witness is now both constructed and quantitative (Proposition 1). (6) KGA's value scales with how often catastrophic, detectable harm occurs—precisely the quantity the theory characterizes; in benign domains always-adapt is strong and the certificate is mainly safety insurance.

a) Claim discipline (what we do and do not assert).:

Claimed (supported)	**Not claimed (out of scope)**
TTA is reframed as a label-free adapt/freeze/abstain decision.	"K-Bound solves TTA."
There is an information-theoretic limit to label-free decisions (Prop. 1).	"K-Bound always beats adaptation."
Under matched evidence a valid certificate must abstain ($\geq 1-2\alpha$, Cor. 1).	"K-Bound is universally optimal."
On our benchmarks KGA cuts harmful adaptation and regret-to-oracle and is robust across regimes.	A single-stream knockout over both trivial policies on every benchmark.

IX. Discussion: when to deploy K-Bound

The theory and experiments converge on a simple deployment rule. K-Bound's value is governed by one quantity—how often *catastrophic, detectable* harm occurs in the deployment stream—and the trichotomy reads off accordingly. **(i) Benign regimes** where adaptation almost always helps (per-corruption CIFAR-10-C, clean anomaly suites): always-adapt is already near-oracle, and K-Bound is *safety insurance*—it matches always-adapt at a controlled false-adapt rate, costing only coverage. **(ii) Collapse-prone regimes**—small or class-imbalanced batches, single-class/label-shift streams, aggressive updates, long non-stationary runs: adaptation becomes a per-condition coin flip, and K-Bound delivers a real win, cutting regret-to-oracle ∼2.5–5.4× versus always-adapt at 0% false-adapt on the CIFAR-10-C stress grid (Table VII). **(iii) Self-protecting candidates** (e.g. SAR's reset): the headroom shrinks, and K-Bound *ties* rather than beats—and never does worse. The practical takeaway: deploy K-Bound as a default guard around any TTA candidate; its overhead is one forward pass, and its benefit scales with exactly the risk it is designed to detect.

X. Conclusion

We reframed label-free adaptation as a decision constrained by identifiability: adapt when benefit is certifiable, freeze when harm is certifiable, abstain when the unlabeled evidence cannot tell. We proved the unknowable regime is fundamental (with a witness), gave a false-adapt-controlled certificate under a stated alignment assumption, proved a covariate-shift positive regime, and reduced sign-of-difference identifiability to an ordinal accuracy judgment on the observable disagreement region (Lemma 1); the unconditional bracketing is resolved negatively by the dichotomy, and what remains open is the weakest falsifiable class under which the one-bit supplement suffices. On real anomaly-routing data the certificate abstains where benefit vanishes and recovers large performance under failure. The CIFAR-10-C Protocol-A re-run supplies the missing catastrophic-harm audit: with every per-condition array serialized across seeds 0–4, the TENT/EATA beats-both claim survives paired bootstrap and Holm correction, while SAR correctly remains a tie when harmful cells are rare.

XI. Reproducibility

Code. All code, configs, and result artifacts are public at an anonymized repository linked from the supplementary material (see its `LICENSE`); the K-Bound paper and scripts live under `docs/research/kbound/`.

One-command reproduction. The CPU/numpy stages—theorem validators plus the anomaly-routing, regression, witness, and ablation experiments—run via `python docs/research/kbound/scripts/99_reproduce_kbound.py -run`. The deep-TTA stress grid runs on a single commodity GPU (MPS or CUDA) via `bash docs/research/kbound/scripts/run_decisive_tta.sh`, which builds the environment, fetches the data, and writes the table and figures.

Environment. Experiments use a `uv/venv` environment, Python 3.12, with the deep-TTA runs pinned to `torch 2.5.1/torchvision 0.20.1`; the CPU-only stages need only `numpy/scipy/scikit-learn/matplotlib`, and `pdflatex` compiles the paper. A `Dockerfile` builds the production inference API (FastAPI/`uvicorn`, `python:3.11-slim` pinned by digest).

Data. CIFAR-10/100 are pulled via `torchvision`; the official CIFAR-10-C corruptions are the Hendrycks–Dietterich set, Zenodo DOI `10.5281/zenodo.2535967` (`CIFAR-10-C.tar`, md5 `56bf5dce`...), auto-downloaded and checksummed by `scripts/fetch_cifar_c.py`.

Determinism and honesty. Single-run stages use fixed seeds; the locked deep-TTA stress grid reports seeds 0–4 with a split-conformal radius at $\alpha = 0.1$. The deep-TTA grid is *pre-registered* (declared before any result is observed) and reported in full, with no condition dropped. Every theorem ships a numerical sanity-check script under `experiments/kbound/theory_validation/`; all were re-run from a clean environment and pass. Result manifests and locked CIFAR artifacts (`result_manifest.json`, `decisive_tta_results.json`, `LOCKED_ANALYSIS_RESULTS.json`, `per_condition_cifar10c_*`) back every reported number.

Appendix

Everything beyond the five theorems lives, with full proofs and per-result validators, in the companion manuscript (`manuscript/`; built from the same validated section sources in `paper/sections/`) and the archived technical report `kbound_full58_backup_2026-06-10.tex`: the RGF reliability-gated-fusion instantiation (T1–T9, GDR); multiclass and regression reductions; the sequential anytime-valid e-process certificate and its β-drift variant; label-shift, bounded- and smooth-drift boundaries unified by the ambiguity reach; router capacity; the multi-candidate route with the γ-meter, the agreement-invisible reach, and the corrected hypothesis H; the budget audit; and the agreement–accuracy (anchored AoL) law. Table II gives the mapping. No result was deleted; each keeps its machine-checked validator (Appendix A). The γ-meter route's CEI diagnostic also has a first *real-data* outcome: on the ImageNet-R track (Table XIII) it fires on all 48 conditions ($\tau \in [1.07, 2.15] \gg \tau^{\star}$=0.52) — co-trained candidates violate conditional error-independence exactly as predicted — and the route abstains rather than certifying falsely, which is the designed behaviour of a checkable guard.

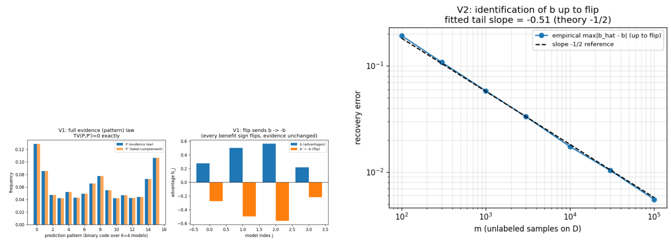

Fig. 11. One-bit identification (Thm 4(ii)): the label complement matches the full evidence law (TV = 0) while negating every advantage (left); recovery of b up to the bit at the $m^{-1/2}$ rate (right).

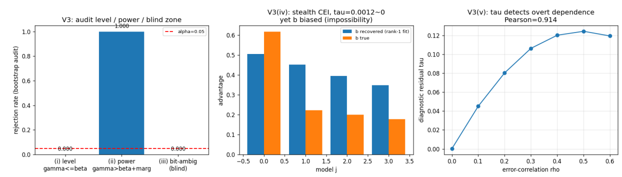

Fig. 12. The budget audit (Thm 4(iv)): sound at level α, power 1.000 beyond the radius, correctly blind exactly on the bit-ambiguous set and the rank-one-preserving stealth law.

A. Validation figures for Theorems 4 and 5

Table XV lists every cell behind the summary in Table VI (KGA wraps Tent; EATA/SAR are the standalone candidates). The per-method means reproduce Table VI exactly. Raw CSV: `experiments/kbound/results/cifar10c_65cells.csv`.

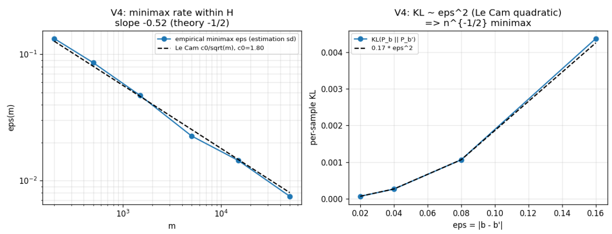

Fig. 13. Evidence-channel rate (Thm 5): empirical minimax error tracks the Le Cam curve $c_0/\sqrt{m}$; KL grows quadratically; the labeled-vs-evidence efficiency ratio diverges as agreement approaches chance.

Fig. 14. Closing Conjecture 1 (Thm 4(iii)): the swap flips every benefit sign while no label-free observable of any order moves (exact zero).

TABLE XV: All 65 CIFAR-10-C cells (13 official corruptions × severities 1–5), ResNet-18, N=800 balanced held-out eval per cell. Online TTA accuracy; KGA wraps Tent. Mean row matches Table VI.

corruption	sev	frozen	Tent	EATA	SAR	KGA	oracle
brightness	1	0.863	0.858	0.856	0.859	0.858	0.865
brightness	2	0.844	0.864	0.864	0.865	0.864	0.868
brightness	3	0.804	0.837	0.837	0.839	0.837	0.839
brightness	4	0.731	0.818	0.818	0.815	0.819	0.816
brightness	5	0.690	0.791	0.794	0.795	0.791	0.799
contrast	1	0.719	0.831	0.831	0.835	0.831	0.831
contrast	2	0.559	0.823	0.821	0.822	0.820	0.820
contrast	3	0.325	0.790	0.793	0.793	0.790	0.791
contrast	4	0.223	0.728	0.729	0.730	0.726	0.726
contrast	5	0.160	0.611	0.607	0.609	0.614	0.601
defocus_blur	1	0.264	0.683	0.680	0.681	0.682	0.676
defocus_blur	2	0.207	0.543	0.544	0.544	0.540	0.540
defocus_blur	3	0.204	0.389	0.386	0.398	0.389	0.385
defocus_blur	4	0.176	0.300	0.300	0.303	0.300	0.305
defocus_blur	5	0.183	0.265	0.264	0.261	0.261	0.275
elastic_transform	1	0.244	0.528	0.528	0.530	0.528	0.525
elastic_transform	2	0.179	0.436	0.440	0.446	0.436	0.452
elastic_transform	3	0.168	0.401	0.401	0.405	0.401	0.400
elastic_transform	4	0.144	0.407	0.407	0.413	0.407	0.410
elastic_transform	5	0.151	0.347	0.346	0.353	0.351	0.351
frost	1	0.725	0.804	0.805	0.806	0.804	0.796
frost	2	0.588	0.760	0.762	0.761	0.761	0.762
frost	3	0.546	0.685	0.685	0.686	0.684	0.697
frost	4	0.492	0.705	0.705	0.704	0.705	0.706
frost	5	0.436	0.640	0.642	0.644	0.640	0.659
gaussian_noise	1	0.487	0.741	0.740	0.741	0.739	0.739
gaussian_noise	2	0.275	0.699	0.699	0.700	0.700	0.701
gaussian_noise	3	0.144	0.594	0.595	0.595	0.593	0.592
gaussian_noise	4	0.100	0.435	0.435	0.432	0.436	0.441
gaussian_noise	5	0.089	0.305	0.304	0.301	0.305	0.301
impulse_noise	1	0.615	0.769	0.770	0.769	0.769	0.775
impulse_noise	2	0.370	0.682	0.682	0.682	0.682	0.688
impulse_noise	3	0.197	0.605	0.605	0.604	0.608	0.623
impulse_noise	4	0.110	0.475	0.479	0.485	0.476	0.484
impulse_noise	5	0.082	0.335	0.335	0.335	0.335	0.331
jpeg_compression	1	0.676	0.730	0.730	0.731	0.730	0.735
jpeg_compression	2	0.639	0.660	0.659	0.661	0.662	0.663
jpeg_compression	3	0.608	0.655	0.654	0.656	0.654	0.658
jpeg_compression	4	0.509	0.573	0.570	0.570	0.573	0.567
jpeg_compression	5	0.460	0.501	0.501	0.501	0.501	0.500
motion_blur	1	0.495	0.782	0.784	0.785	0.784	0.783
motion_blur	2	0.365	0.666	0.665	0.664	0.666	0.675
motion_blur	3	0.279	0.575	0.574	0.575	0.574	0.590
motion_blur	4	0.255	0.490	0.491	0.489	0.491	0.502
motion_blur	5	0.225	0.456	0.450	0.446	0.456	0.476
pixelate	1	0.429	0.719	0.720	0.721	0.716	0.720
pixelate	2	0.373	0.651	0.651	0.654	0.650	0.654
pixelate	3	0.306	0.570	0.574	0.576	0.573	0.571
pixelate	4	0.267	0.487	0.485	0.487	0.489	0.486
pixelate	5	0.289	0.405	0.404	0.406	0.405	0.410
shot_noise	1	0.492	0.779	0.776	0.779	0.777	0.784
shot_noise	2	0.254	0.654	0.655	0.659	0.654	0.669
shot_noise	3	0.155	0.536	0.537	0.549	0.534	0.545
shot_noise	4	0.125	0.400	0.398	0.398	0.398	0.404
shot_noise	5	0.120	0.303	0.305	0.306	0.300	0.300
snow	1	0.770	0.782	0.781	0.779	0.782	0.781
snow	2	0.578	0.686	0.685	0.685	0.685	0.690
snow	3	0.661	0.716	0.716	0.711	0.718	0.710
snow	4	0.574	0.683	0.681	0.679	0.683	0.680
snow	5	0.459	0.597	0.596	0.597	0.597	0.598
zoom_blur	1	0.760	0.877	0.879	0.876	0.876	0.879
zoom_blur	2	0.708	0.847	0.847	0.849	0.847	0.853
zoom_blur	3	0.674	0.830	0.829	0.828	0.829	0.826
zoom_blur	4	0.635	0.809	0.810	0.810	0.807	0.814
zoom_blur	5	0.557	0.779	0.776	0.778	0.780	0.785
mean		**0.412**	**0.626**	**0.626**	**0.627**	**0.626**	**0.629**

Theorem 1 is proved with a constructed witness and strengthened to a quantitative Le Cam minimax floor (part (ii)); Lemma 2 is a plug-in regret decomposition (exact identity) with a minimax floor; Theorem 2 is proved *conditional* on a valid label-free interval estimator and complemented by an anytime-valid e-process (manuscript, App. B); Corollary 1 proves the covariate case and Lemma 1 the binary, multiclass, and regression sign-of-difference results. The multiclass/regression label-free *bracketing* is closed under a bounded calibration-transfer assumption (manuscript, App. C); the *unconditional* bracketing (Conjecture 1) is resolved as a testability dichotomy (Theorem 4: no evidence-definable assumption identifies the sign; the minimal untestable supplement is one bit, which suffices under falsifiable structure). Every theorem ships a numerical sanity-check script (`experiments/kbound/theory_validation/`); all were re-run from a clean environment and pass: the Le Cam floor tracks the closed form, the regret identity holds to $\sim 10^{-17}$, the multiclass identity to 10^{-16} with 100% sign agreement over 4000 trials, calibration-transfer sign-recovery is 100% when the confidence gap exceeds 2η, the router generalization gap decays as $\sim n^{-0.66}$, and the drift-corrected anytime false-adapt rate stays $\leq \alpha$. Experiments report multi-seed means $\pm$ std with paired t-tests, evidence/α/batch ablations, a regression covariate-shift track, and a clean non-identifiability witness.

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